Citrus Twist Assembly Manual

Revision 1.0

DISCLAIMER

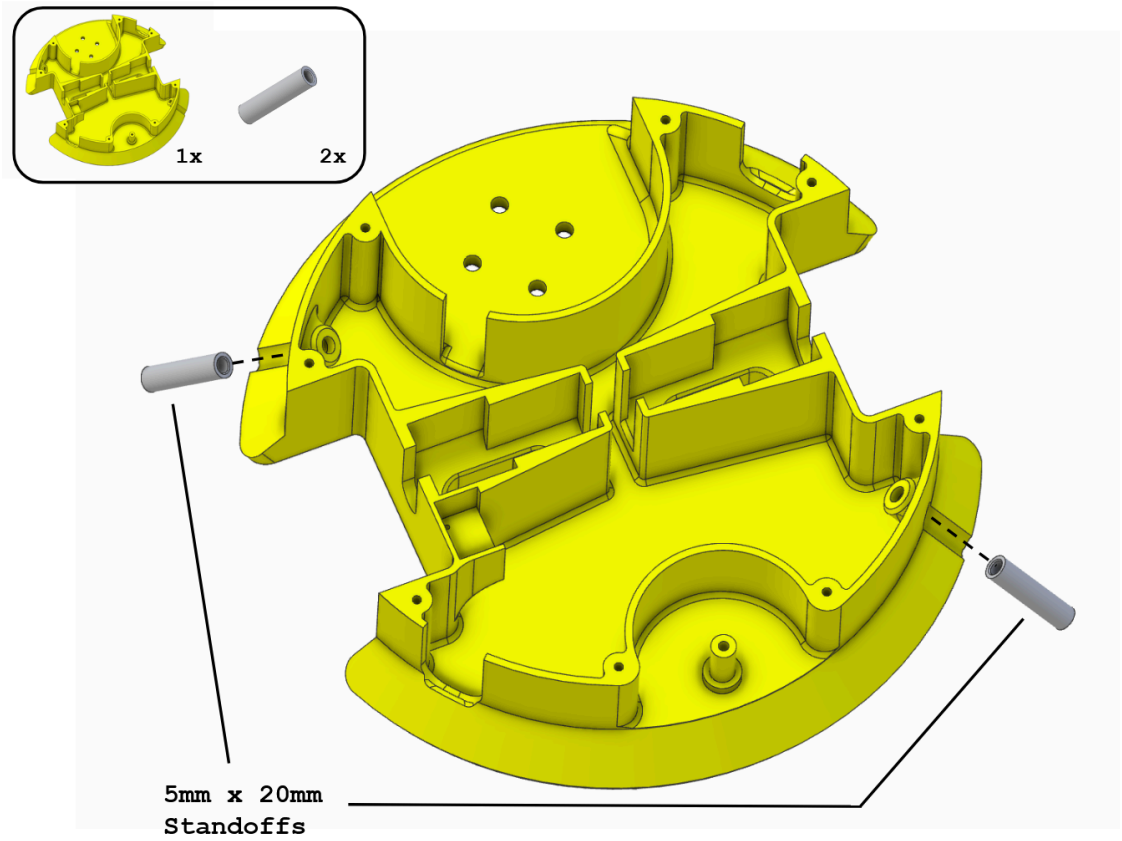
Combat robots are dangerous. It is your responsibility to follow best safety practices and take the proper steps to protect yourself and the people and property around you, including (but not limited to) using a test box and proper weapon lock.

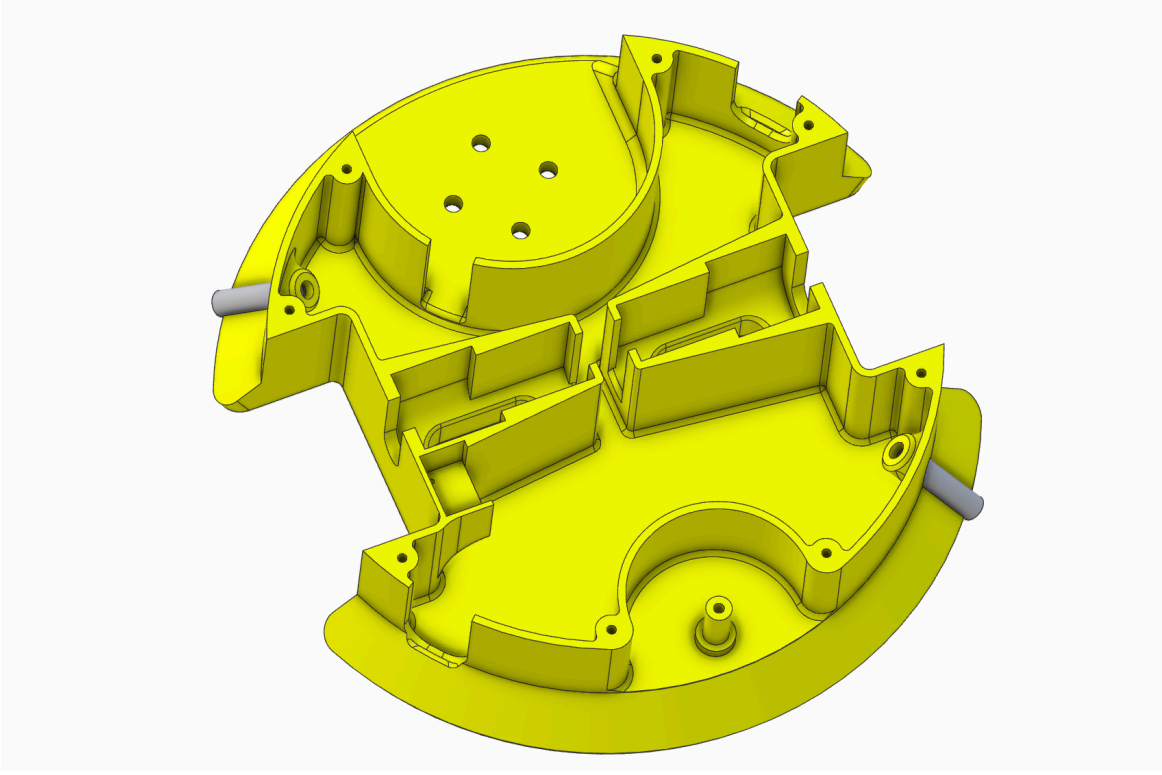
The manufacturers, designers, and distributors of the components and printed parts used in this kit are not responsible for your safety and not liable for any damage or harm caused by the use, proper or improper, of this device.

Be safe, be smart, be responsible, and have fun!

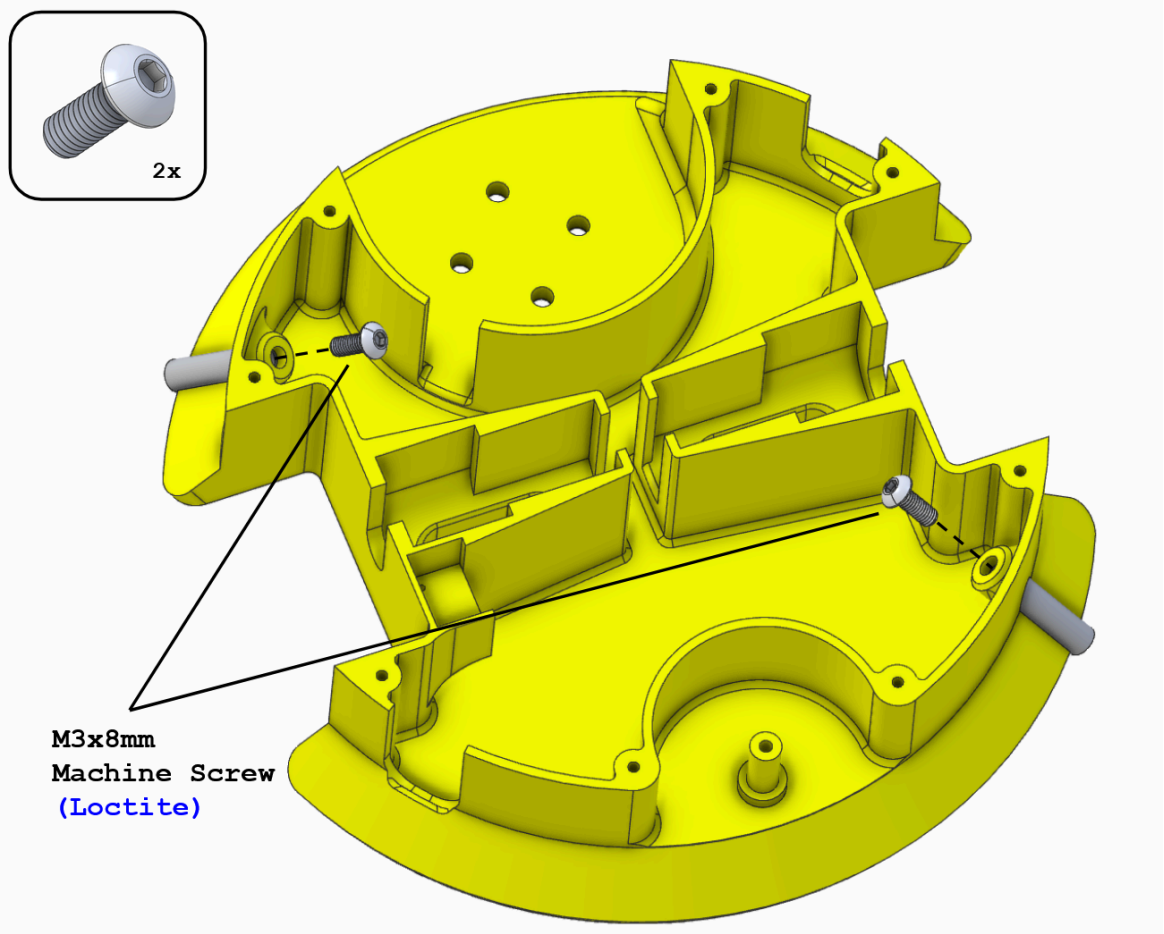
A. Lower Core Assembly

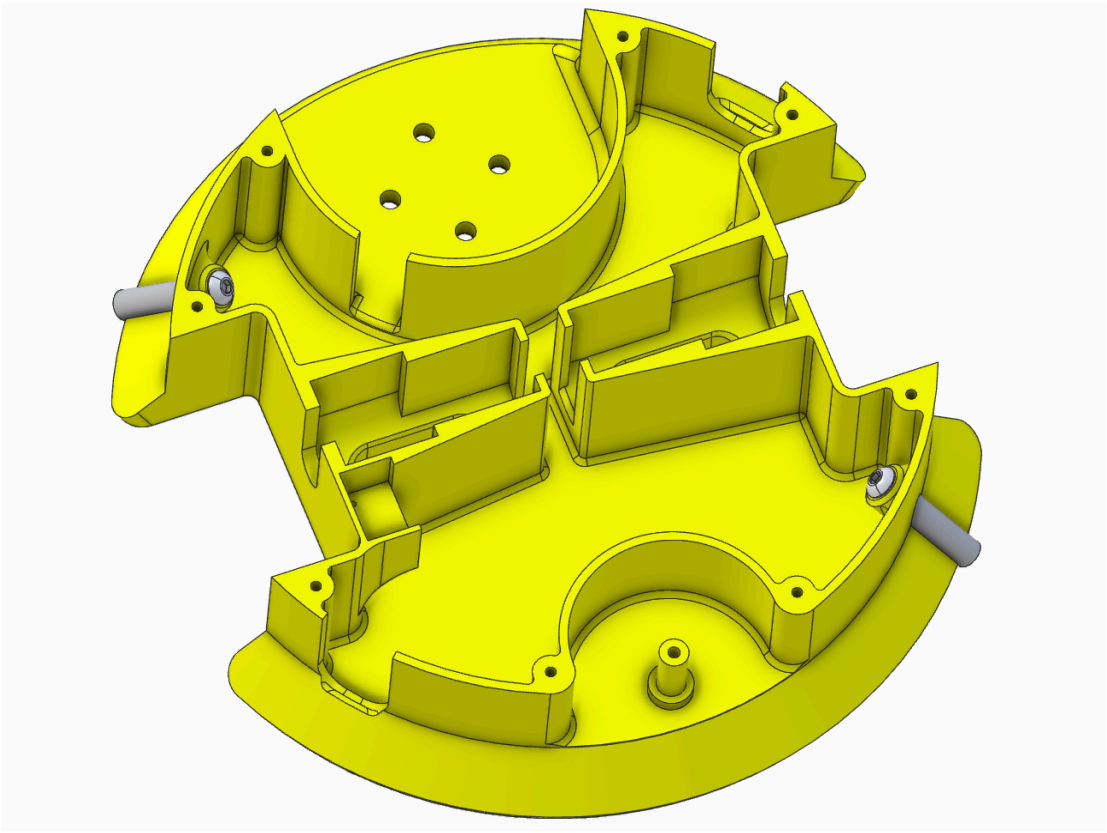
1. Insert 5mm x 20mm standoffs into locating grooves.



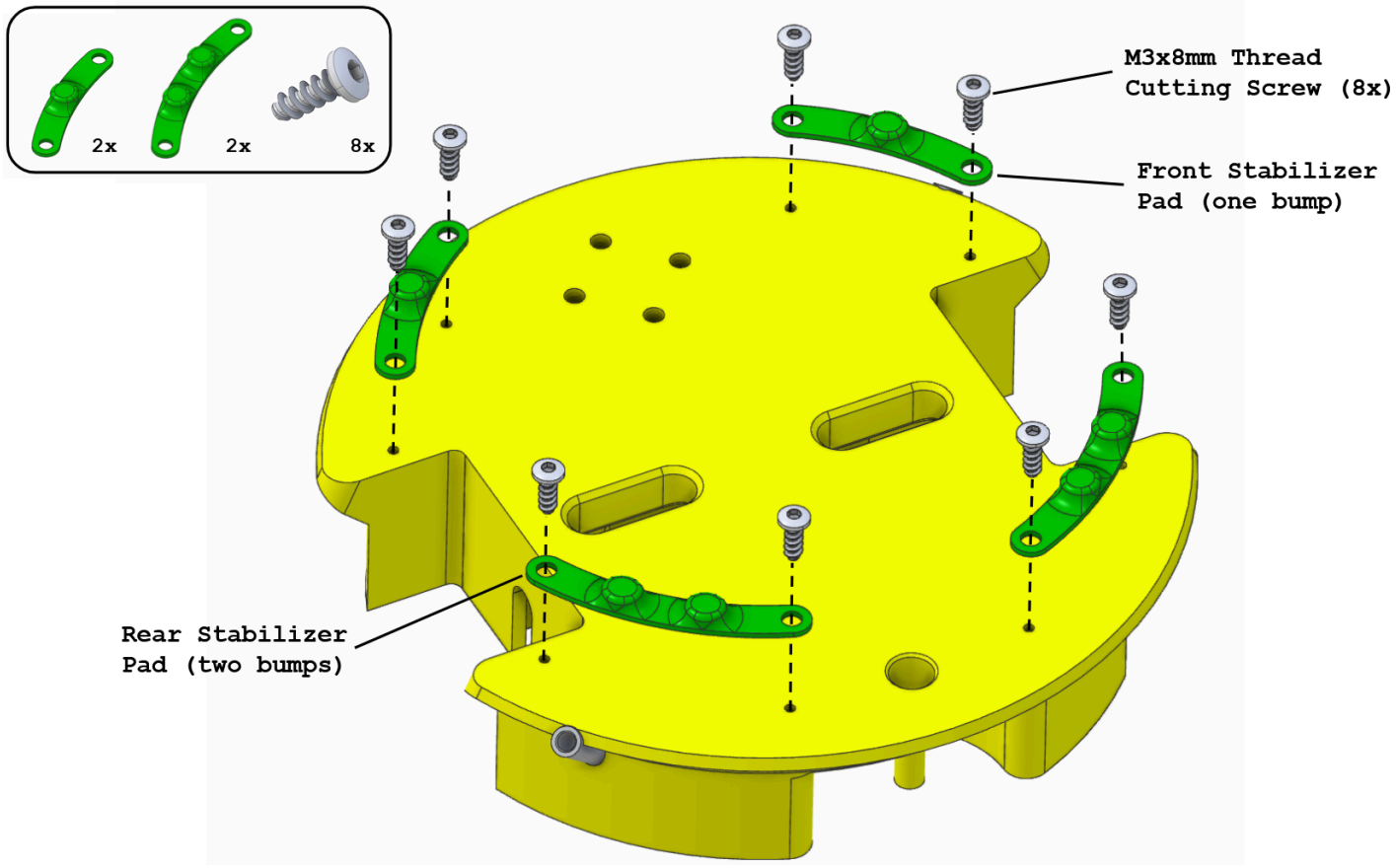


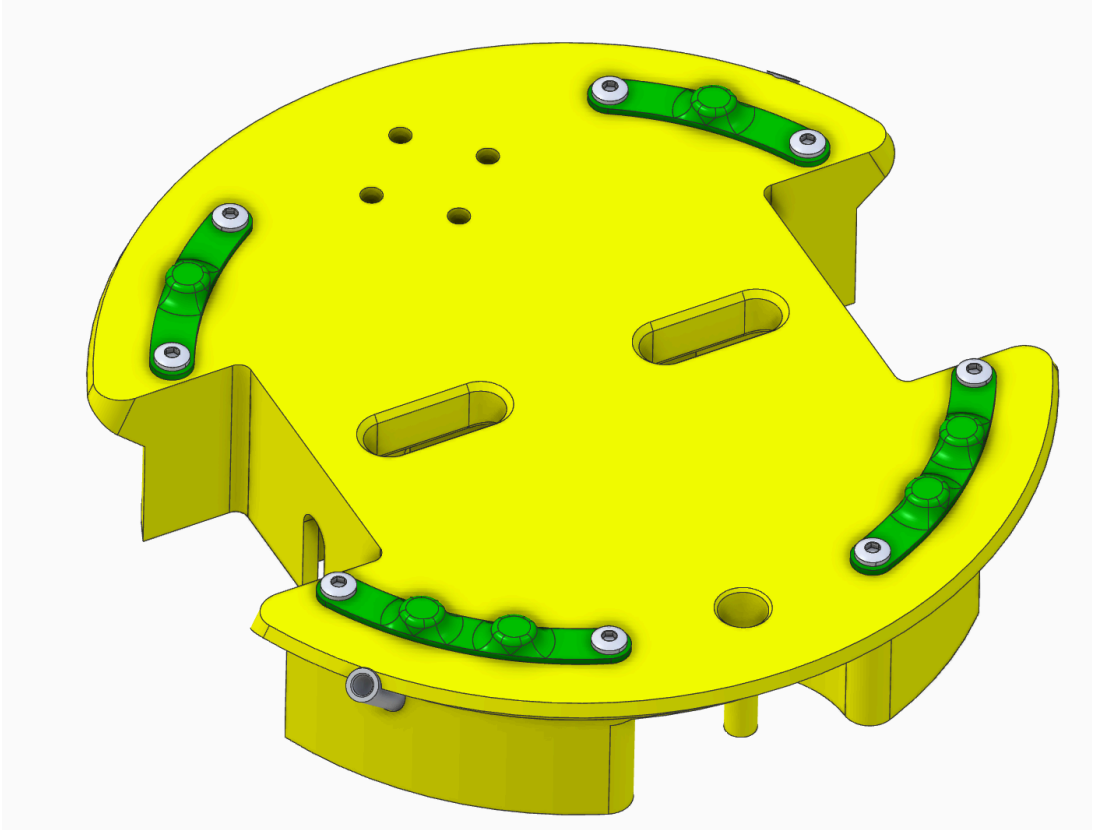
2. Attach guide standoffs with M3x8mm machine screws.



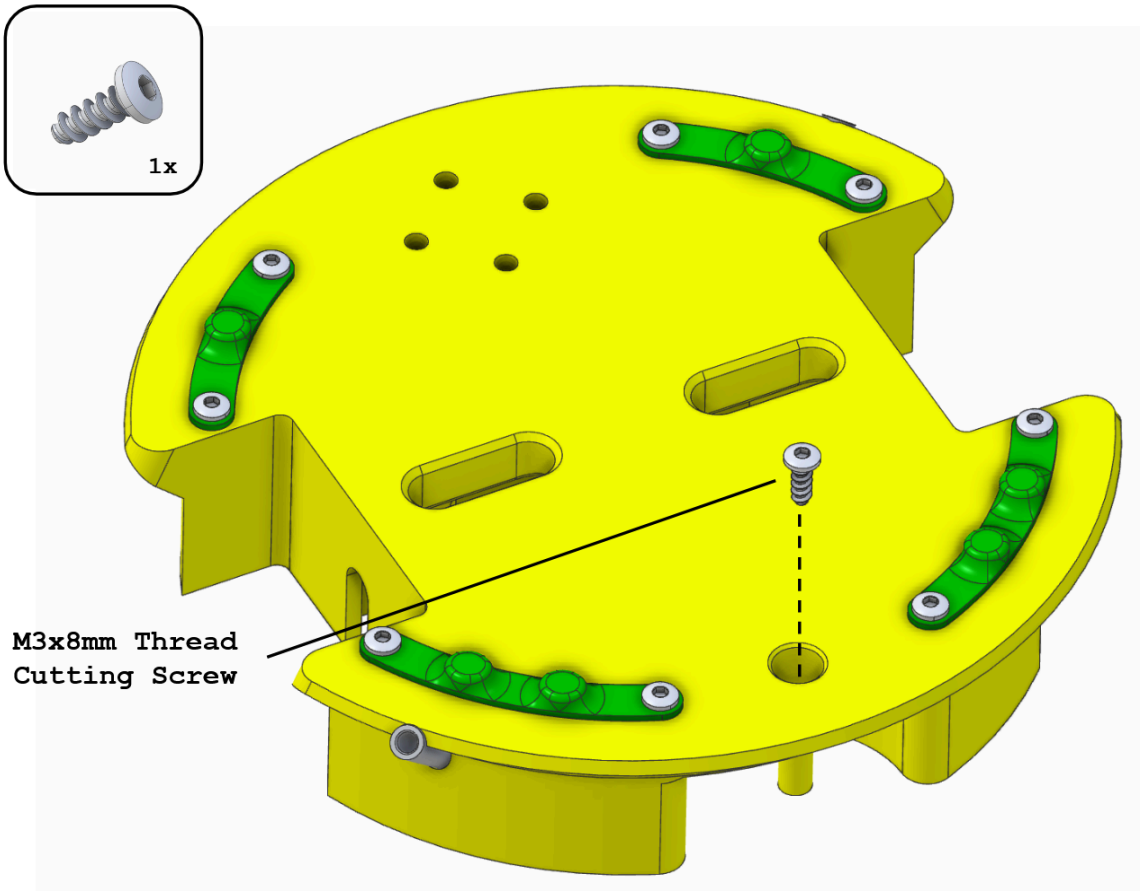


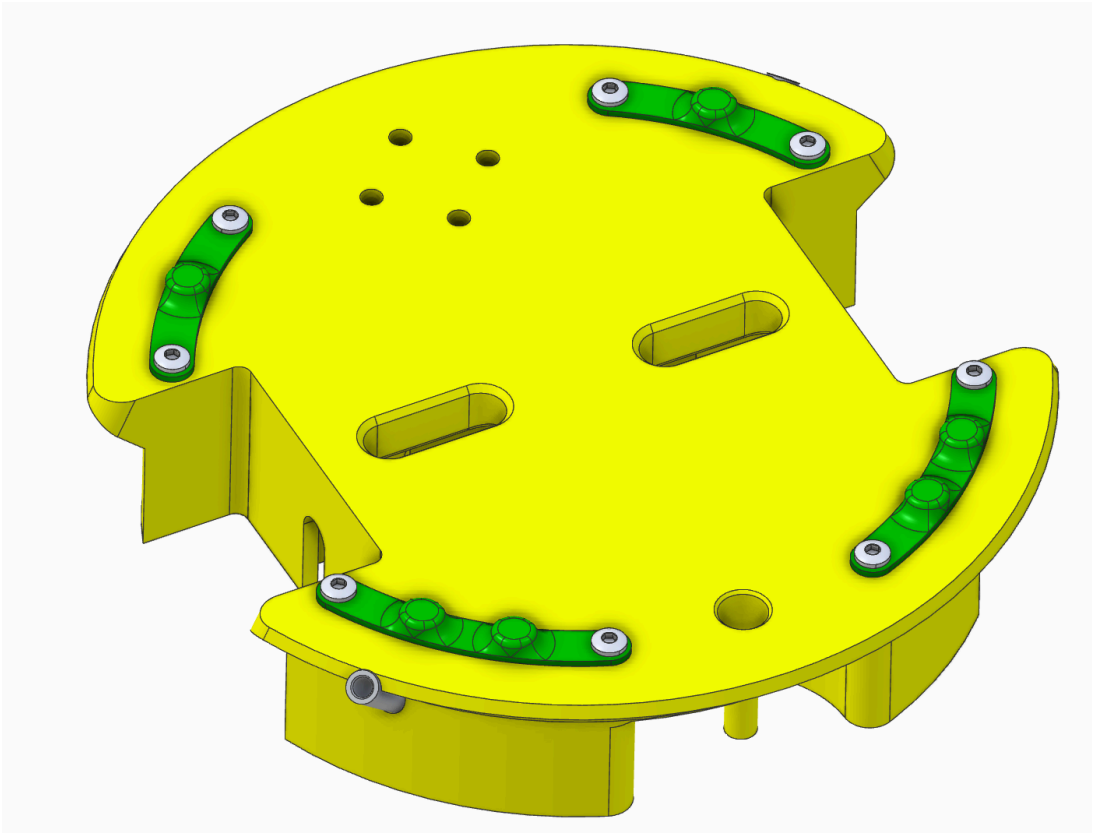
3. Flip body over, attach one set of stabilizer pads with M3x8mm thread cutting screws.



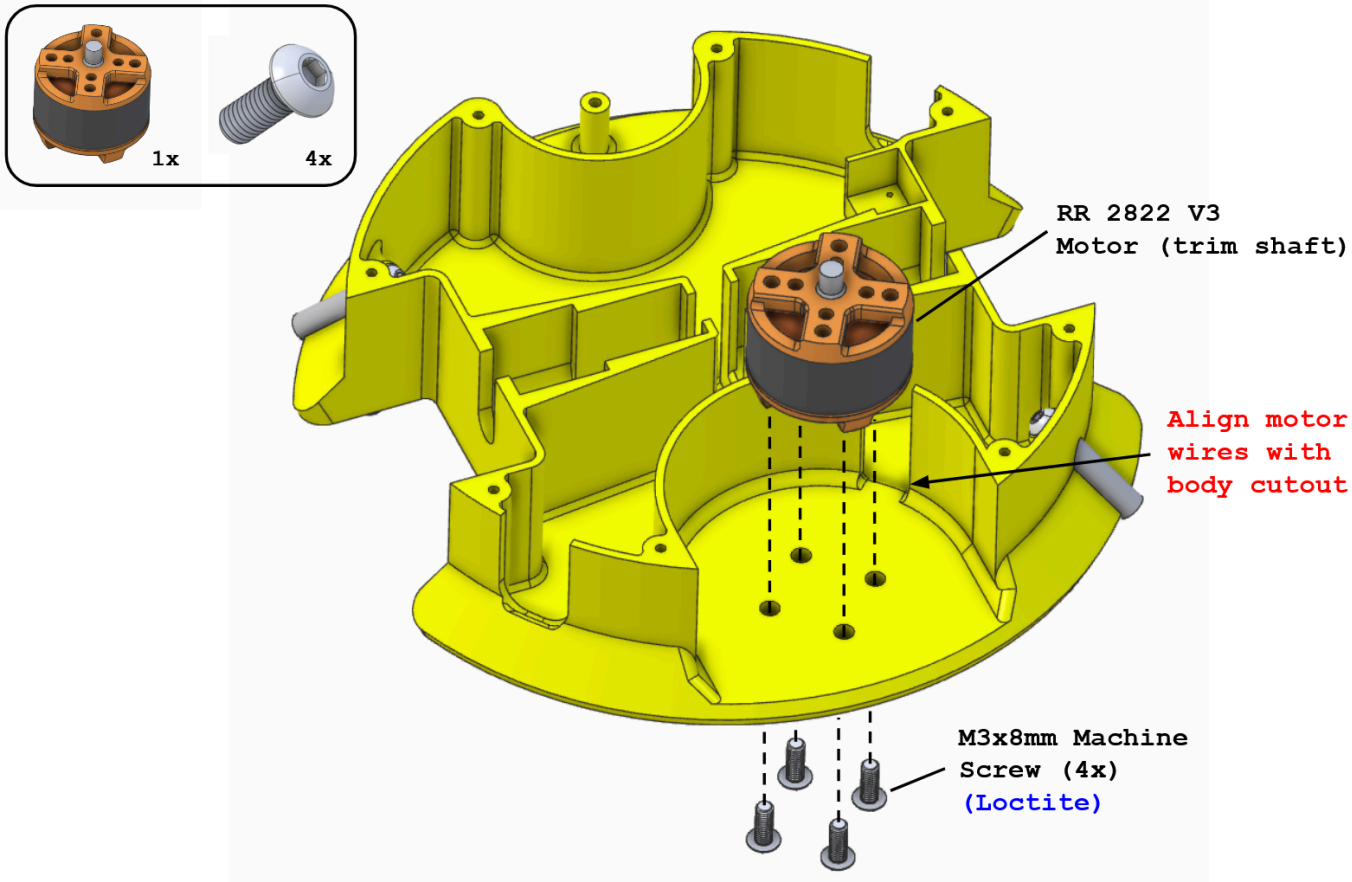


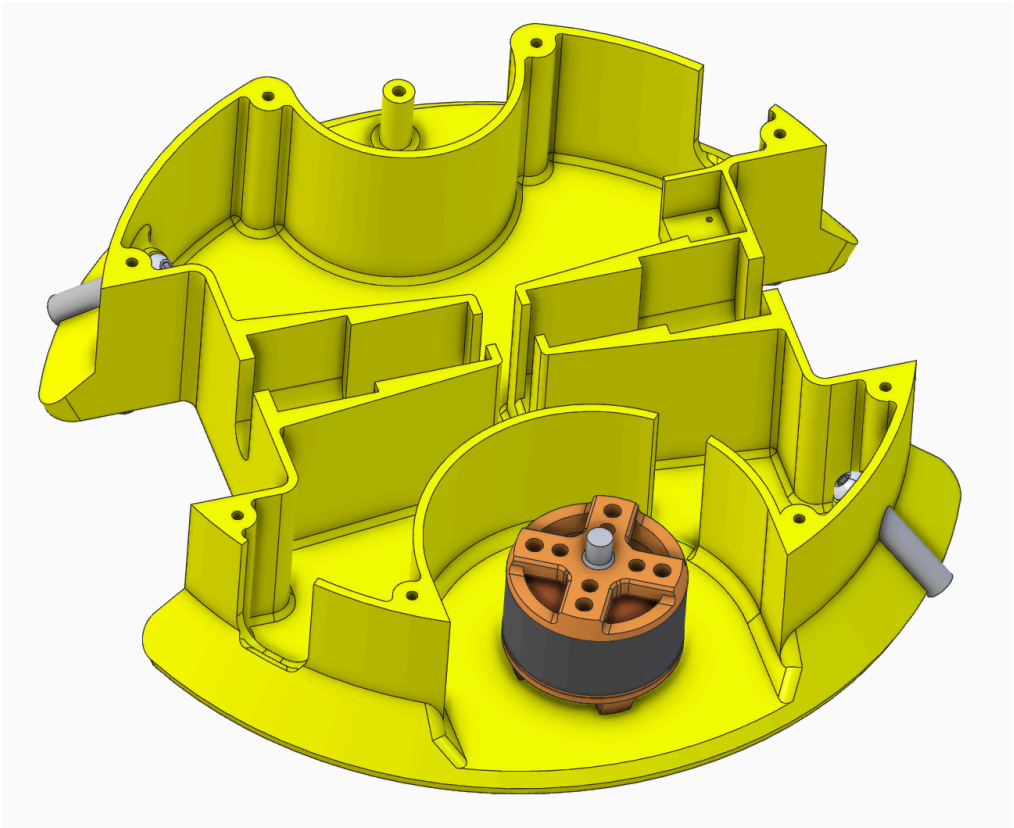
4. Add M3x8mm thread cutting screw to the idler pulley axle post.



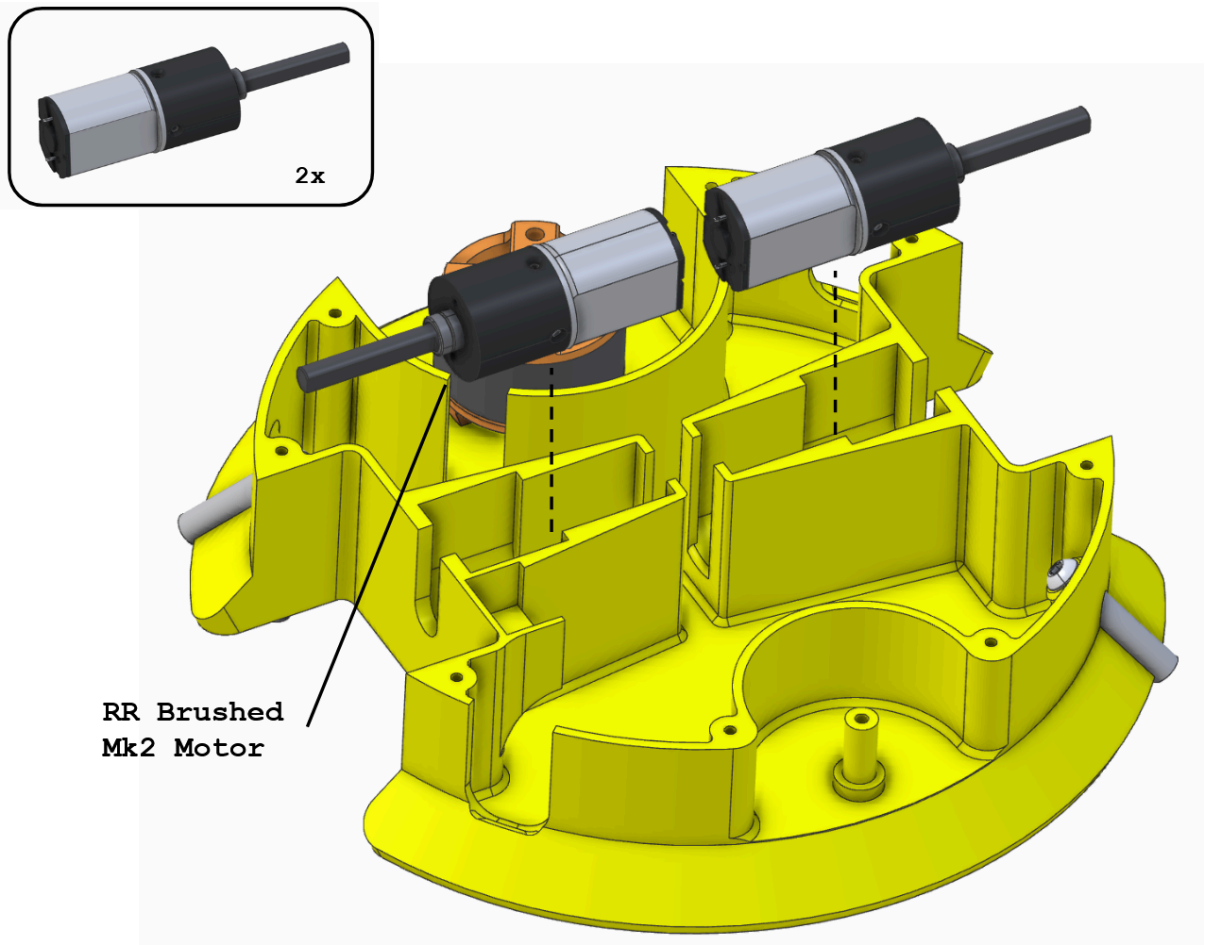


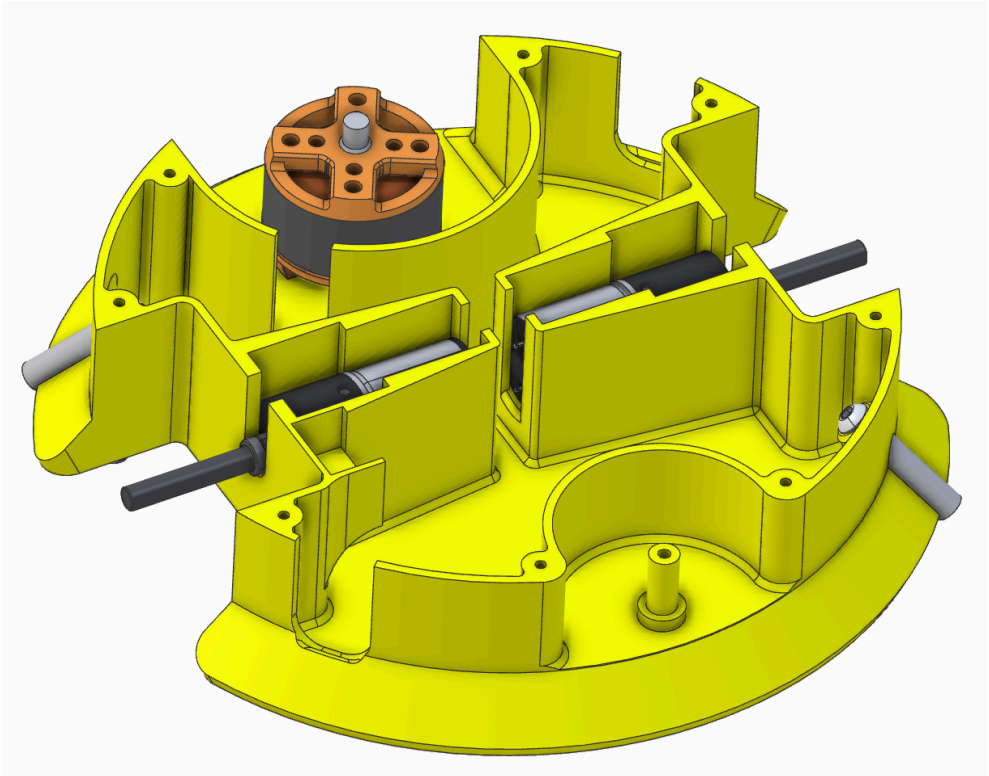
5. Flip back over, attach the RR 2822 V3 weapon motor with M3x8mm machine screws.



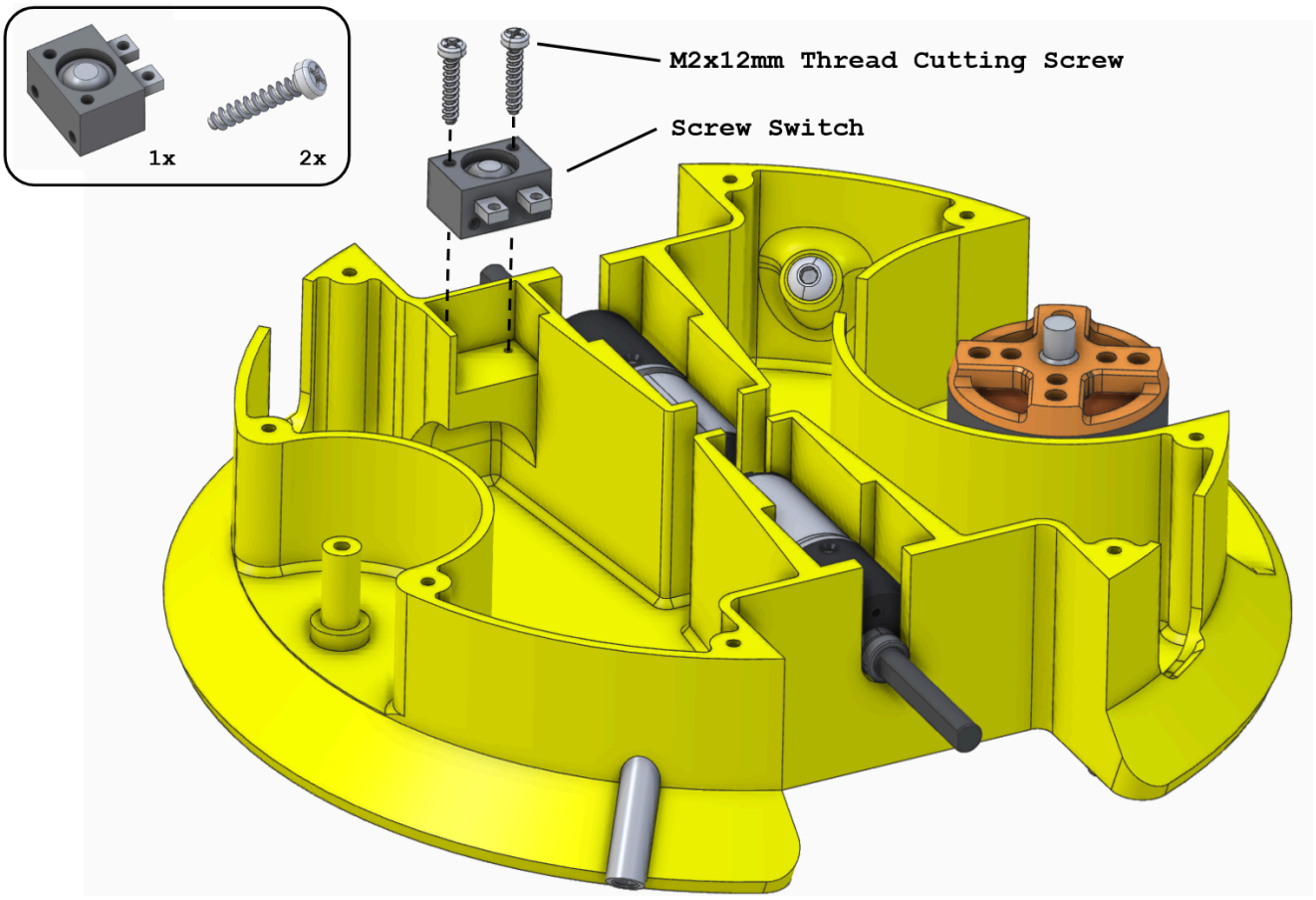


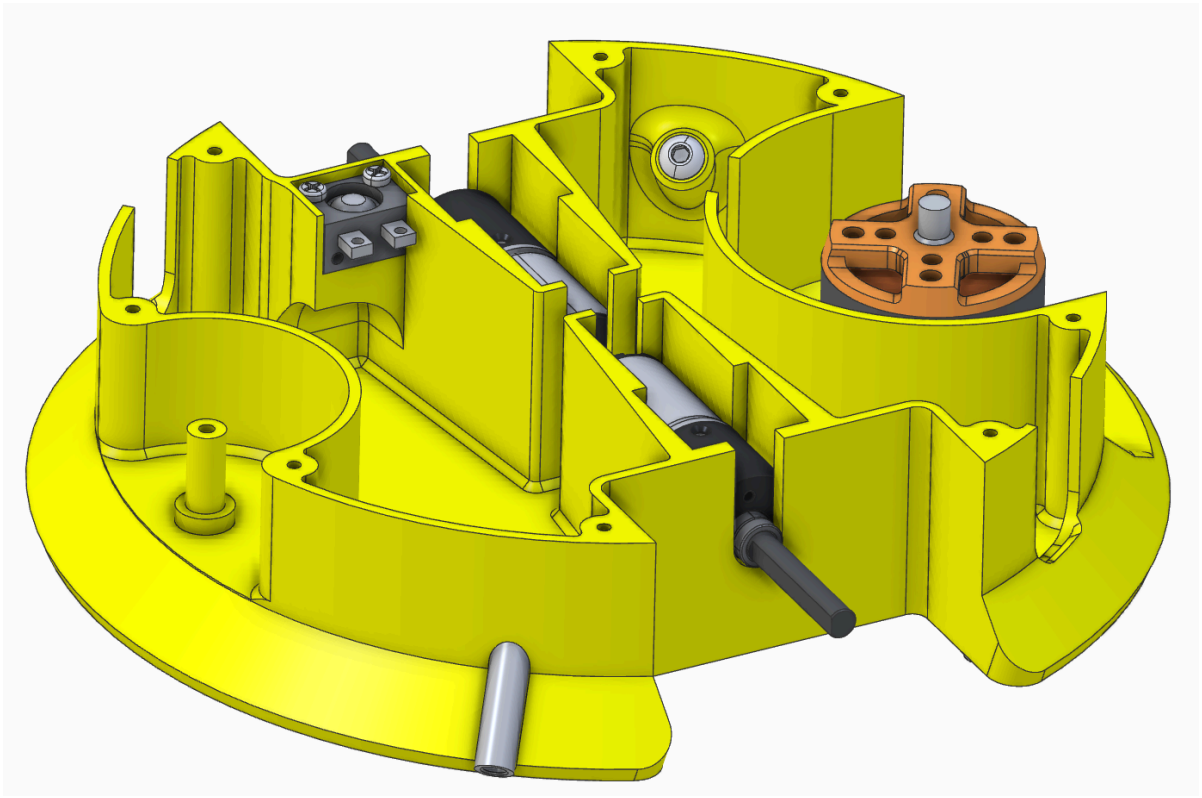
6. Slide the RR Brushed Mk2 drive motors into the retaining slots.





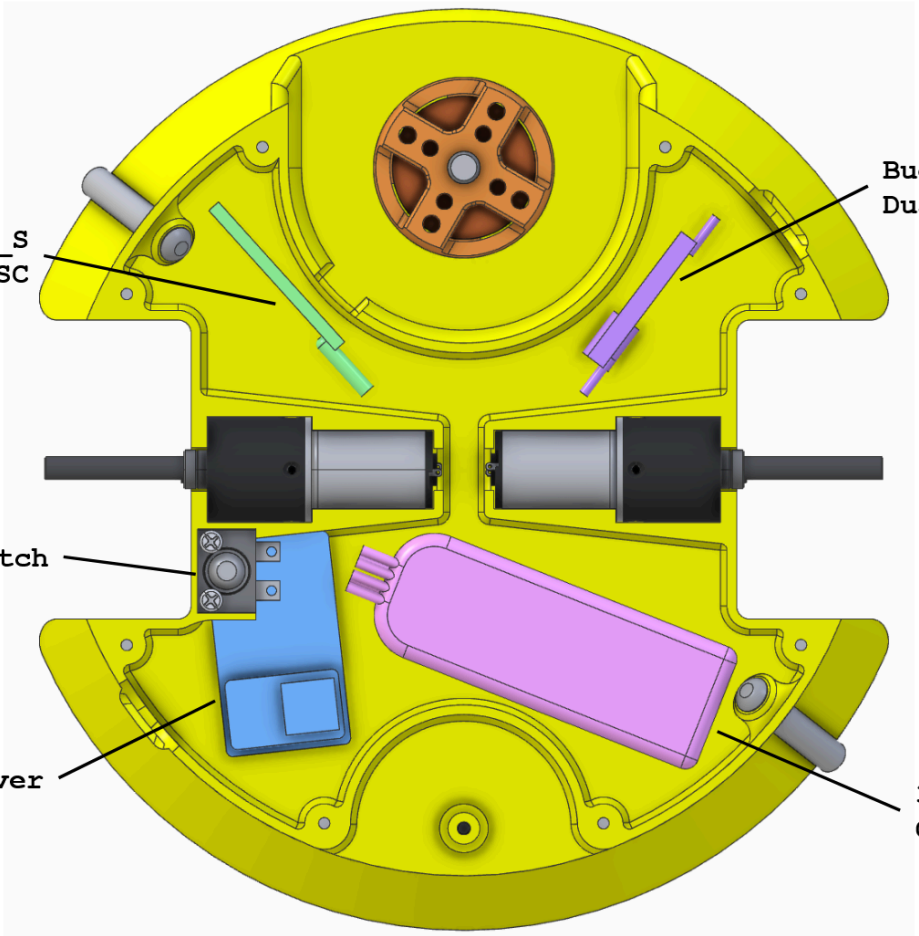
7. Attach screw switch with M2x12mm plastic threading screws.



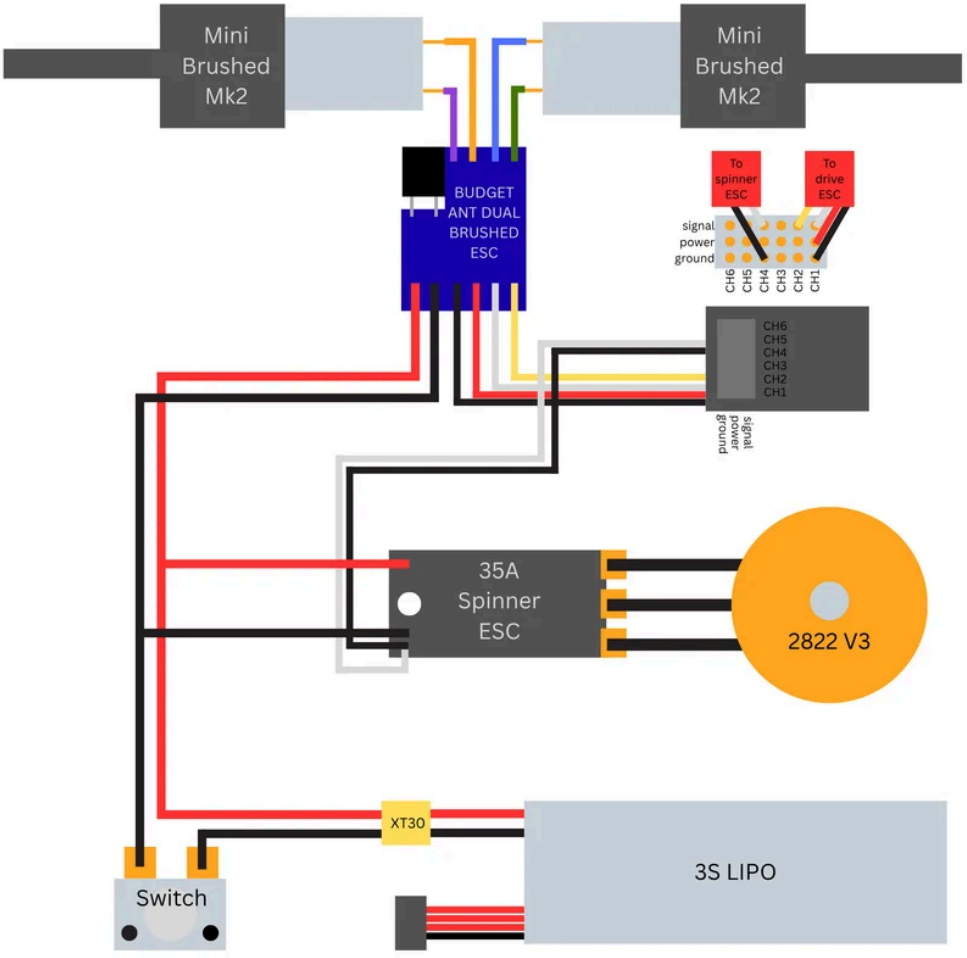


8. Wire your robot!

It's easiest to wire the robot at this point and test the motors are working before anything dangerous is attached. Since your motors and switch are in, use them to measure the wire lengths you'll need and trim them to size. The recommended electronics layout is shown below, but just fit things in wherever you find space. Use the gap between the drive motors to pass wires from the front to back of the robot.

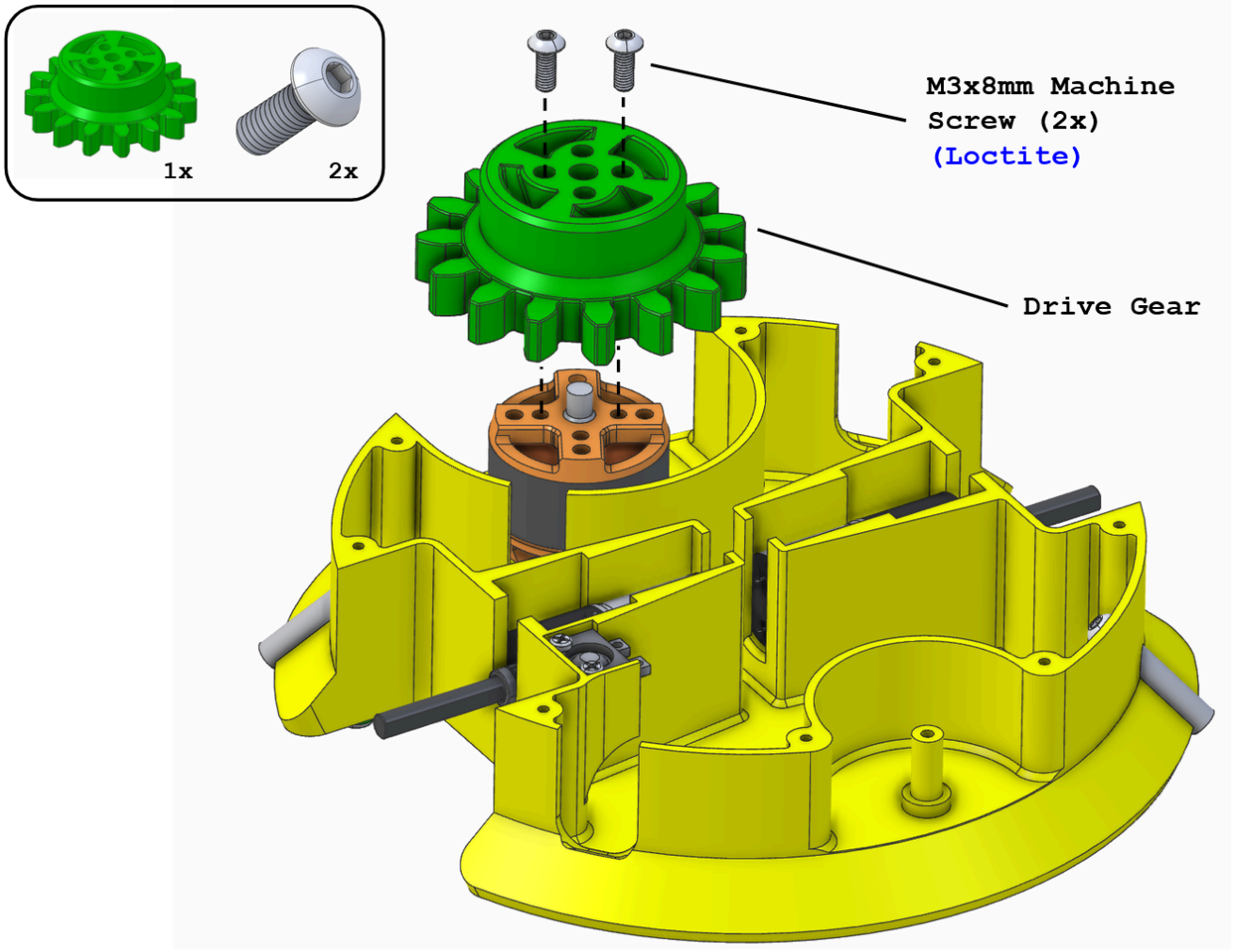


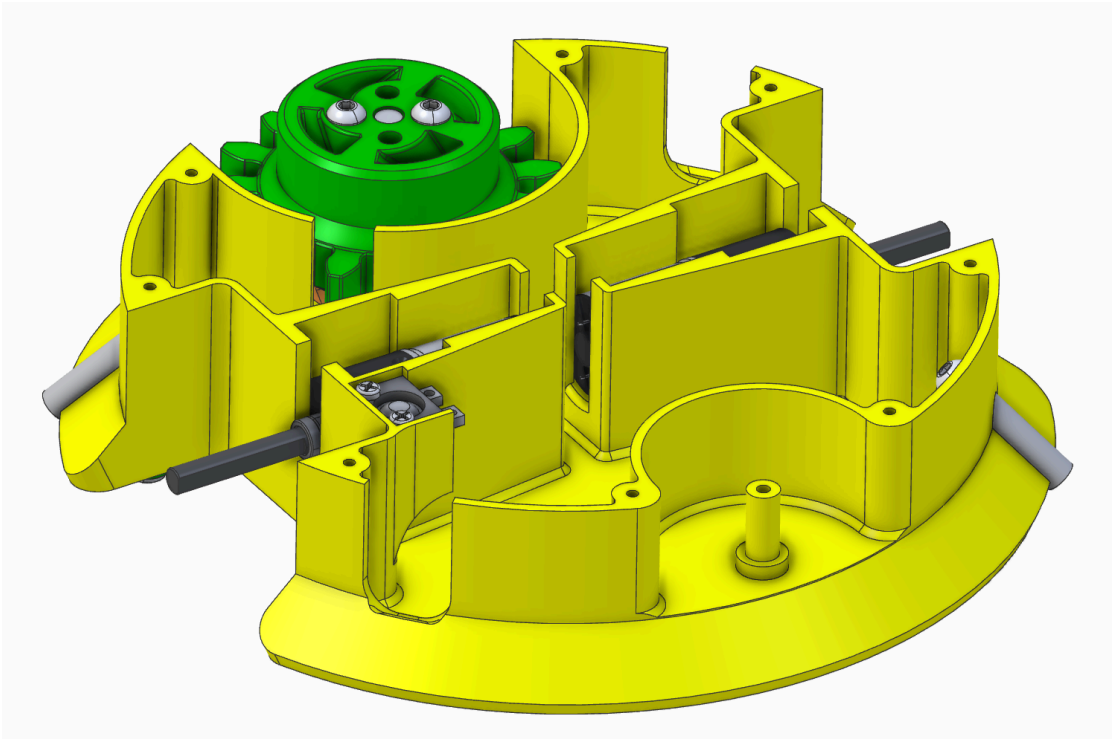
Reference the Repeat Open PLAnt Bundle wiring diagram to make sure you make the right connections when soldering. It is recommended to remove the drive motors prior to soldering to the tabs.



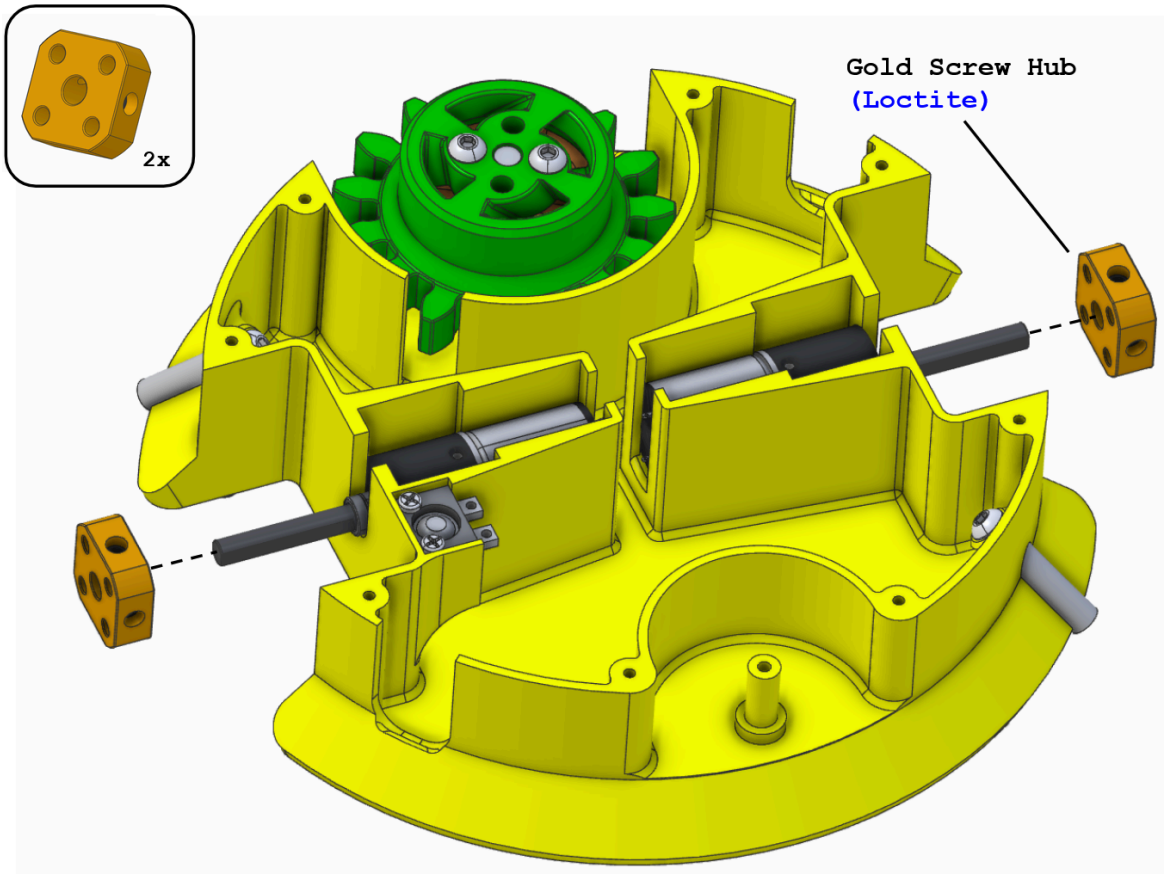
After double-checking and testing the wiring, remove the battery from the robot for safety. Once you've finished the instructions, you can remove the top plate and install it again for testing. The electronics you just installed will not be shown in the remaining steps for simplicity.

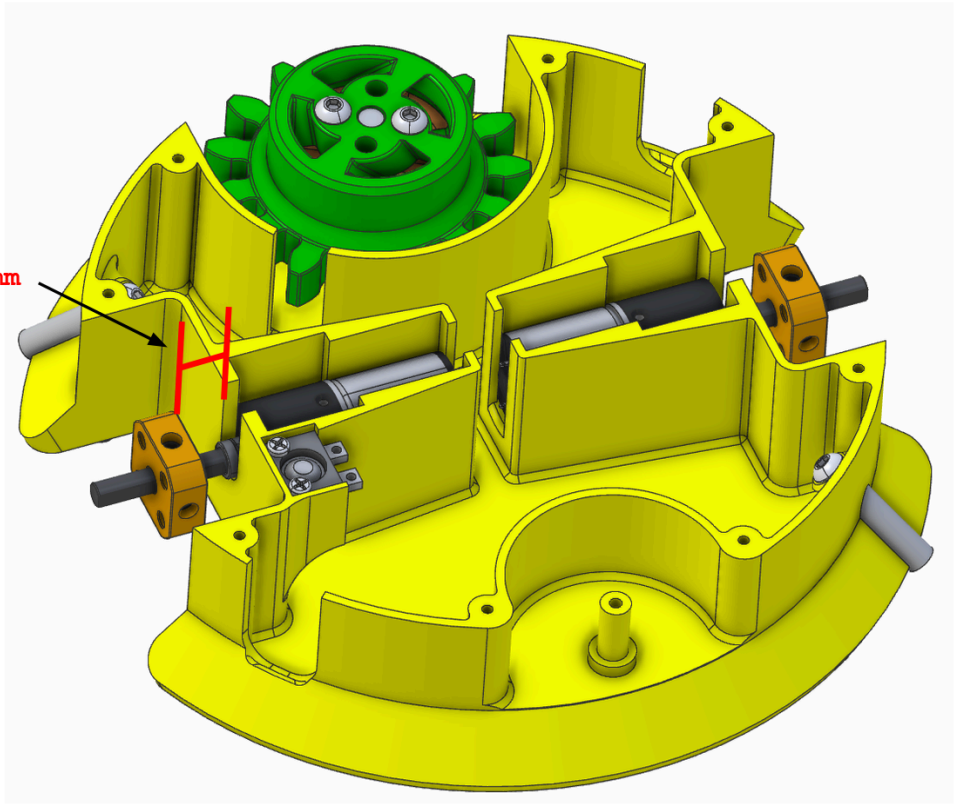
9. Attach the drive gear to the weapon motor with M3x8mm machine screws.



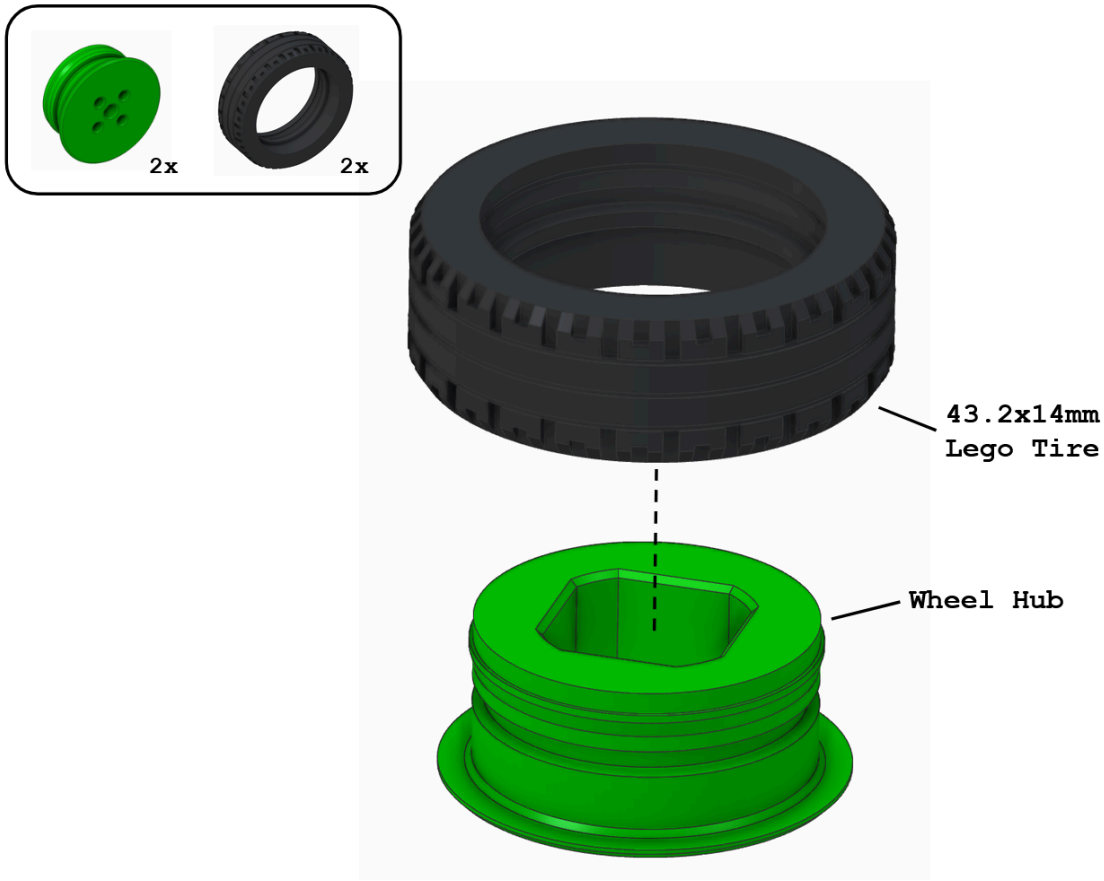


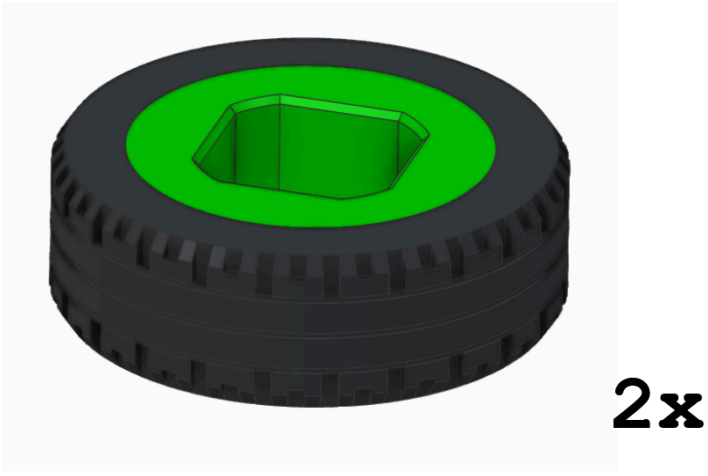
10. Slide the Gold Screw Hubs onto the drive motors and secure them with the setscrews. Leave ~7mm of gap between Gold Screw Hubs and printed body.



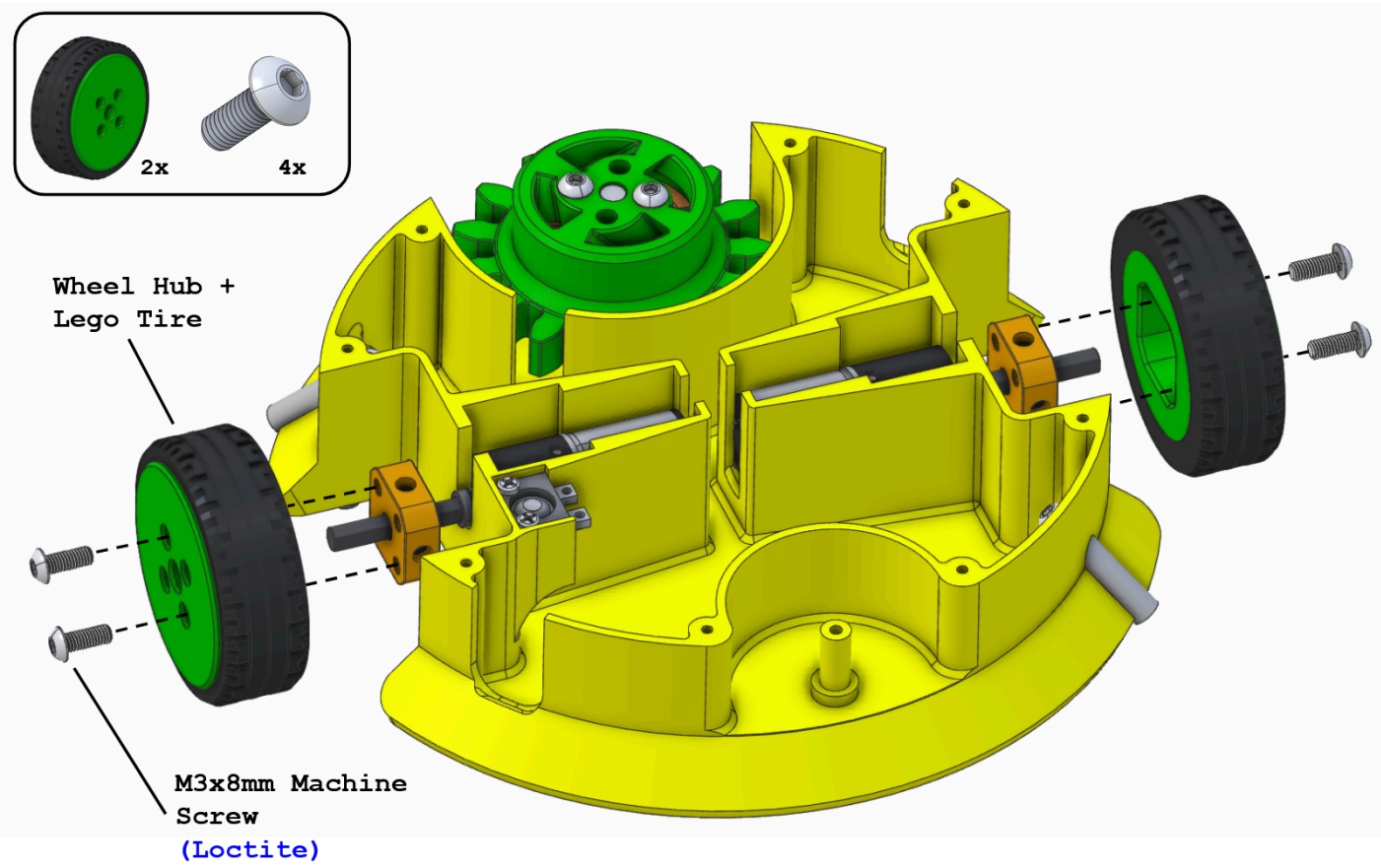


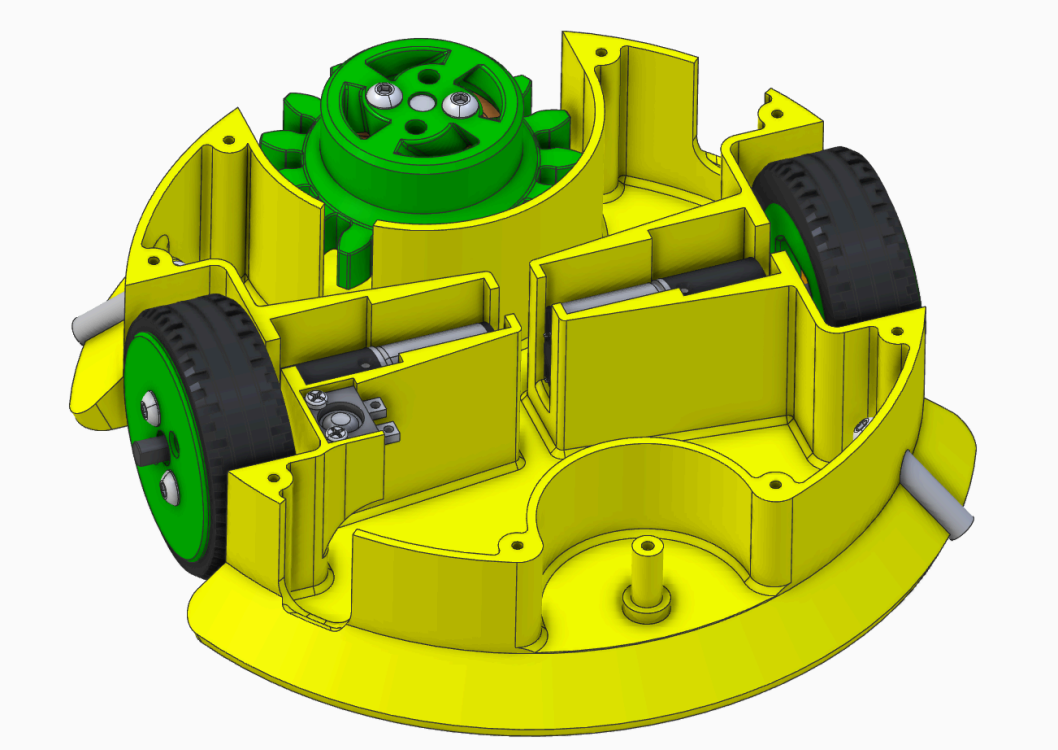
11. Press the 43.2x14mm Lego tires onto the printed wheel hubs.





12. Attach the wheel hubs + tires to the Gold Screw Hubs with M3x8mm machine screws. If tires are rubbing on the body, adjust the screw hub spacing in the prior step.



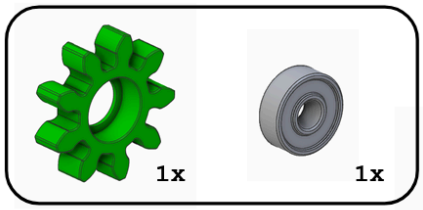


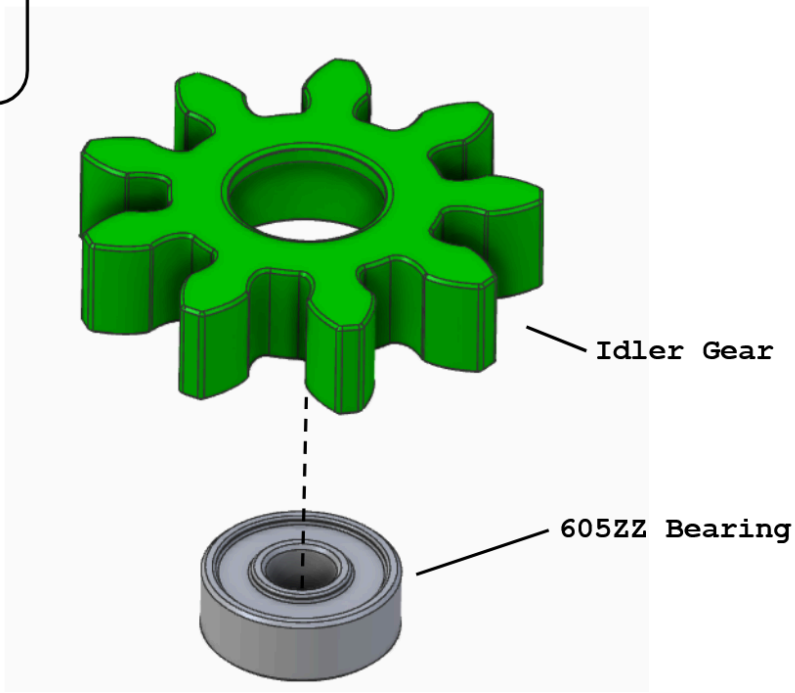
B. Weapon Assembly

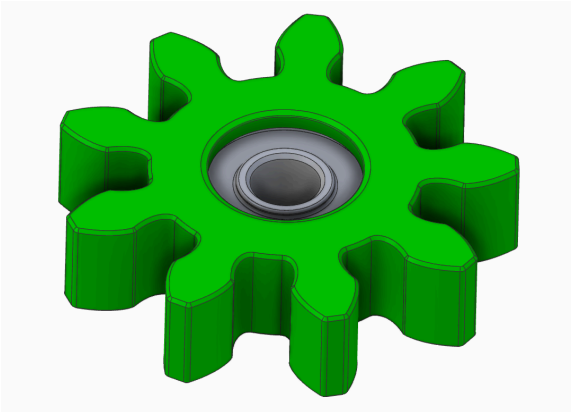
The weapon assembly can be put together without having to unscrew the weapon motor or drive gear, but requires the following setup.

DO NOT run the weapon when the upper core is not attached, as the weapon will not be fully constrained and may damage other parts.

1. Press the 605ZZ bearing into the idler pulley. Note the lip on one side of the idler gear.

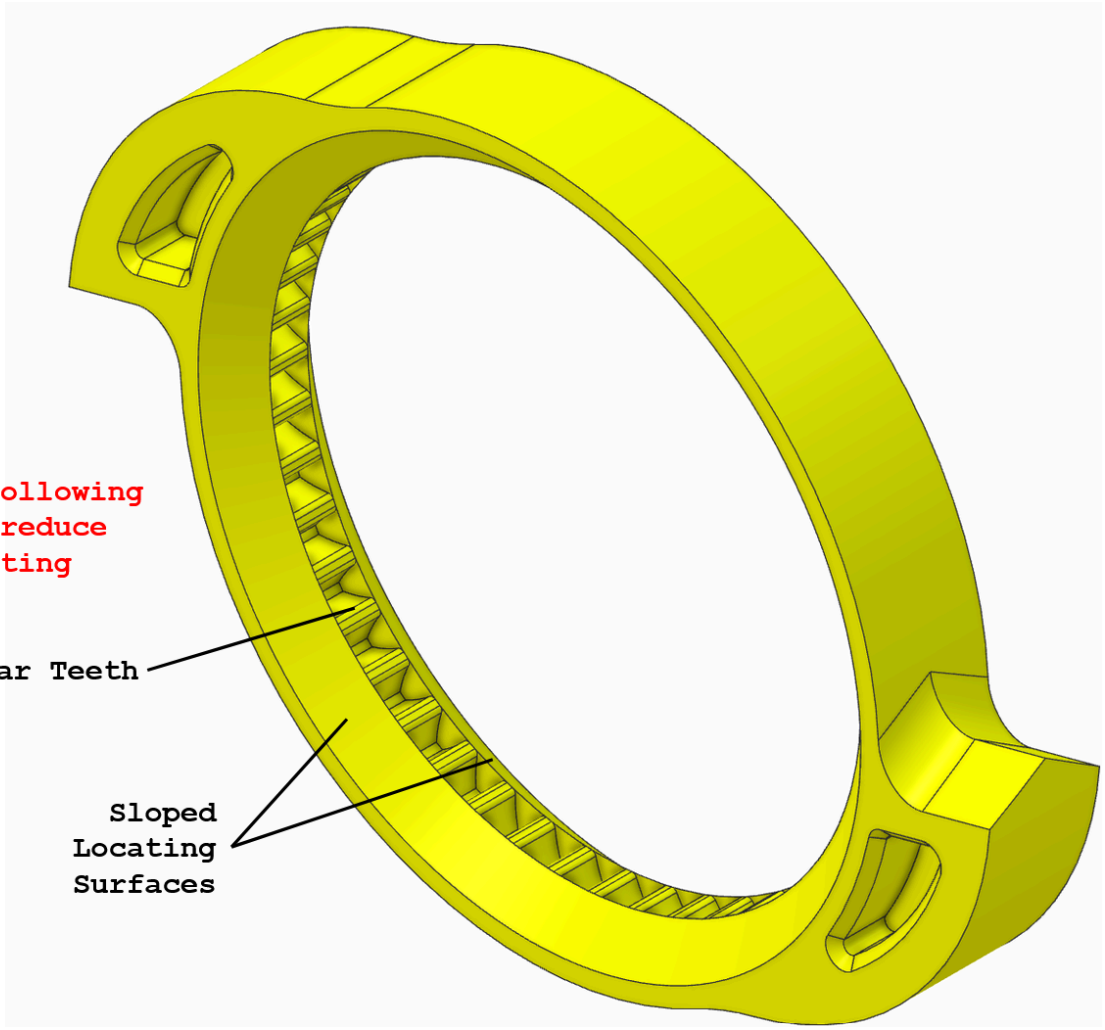




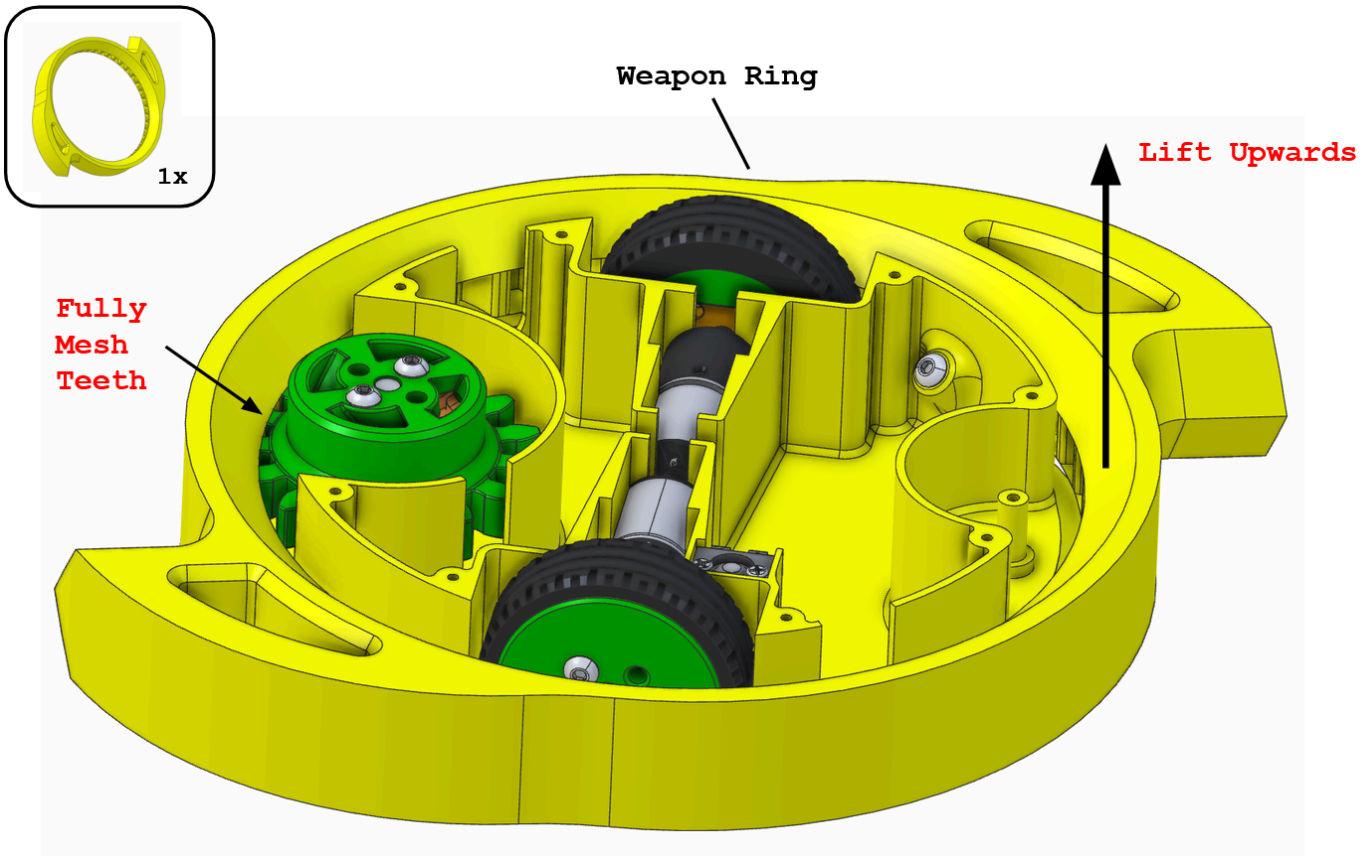


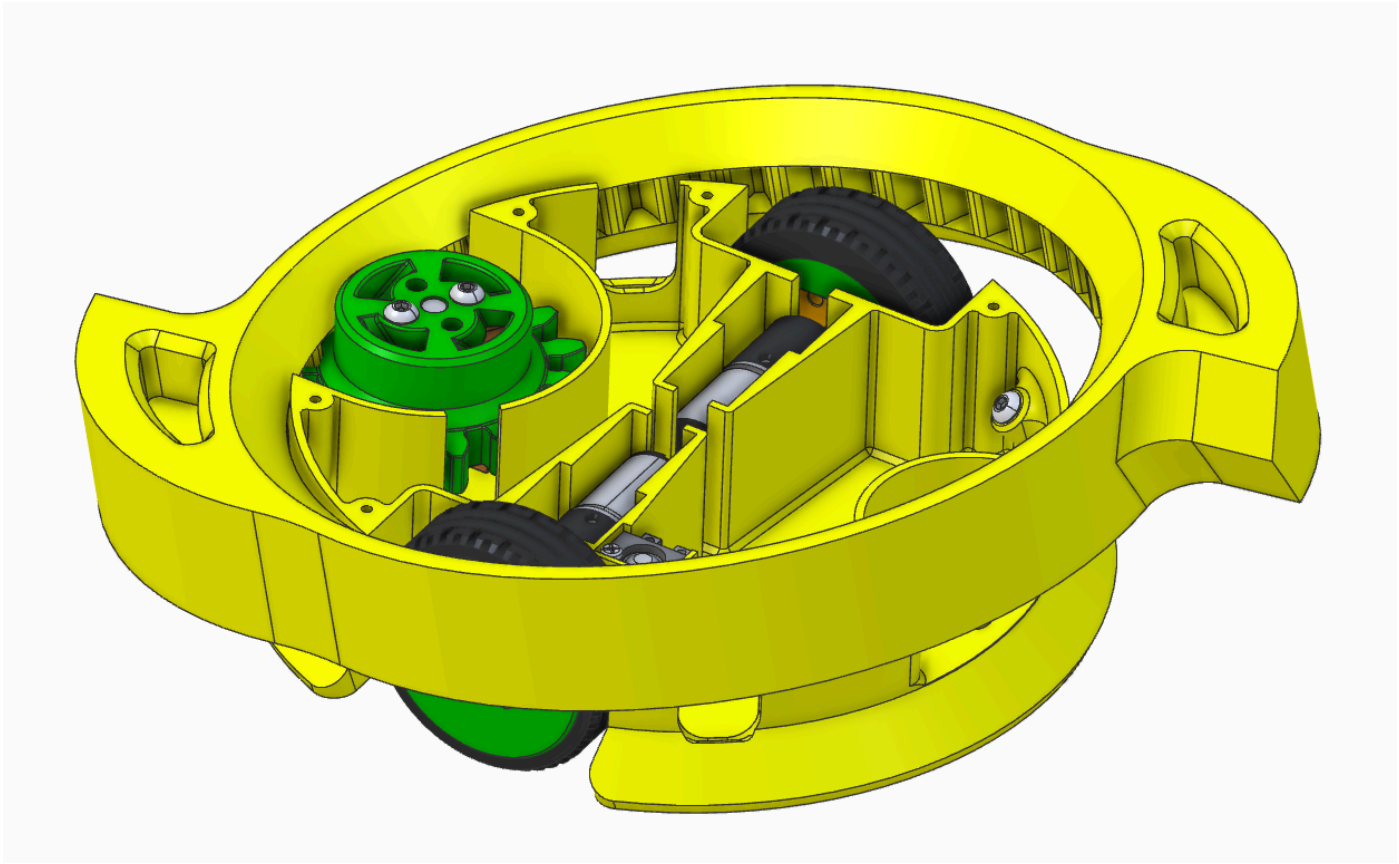
2. Grease weapon ring to prevent melting.

It is highly recommended to add a small amount of grease to the contact surfaces shown below to prevent heat buildup from friction. White lithium grease has been tested and found to work for this application, although other greases/friction reducing compounds may also be viable.

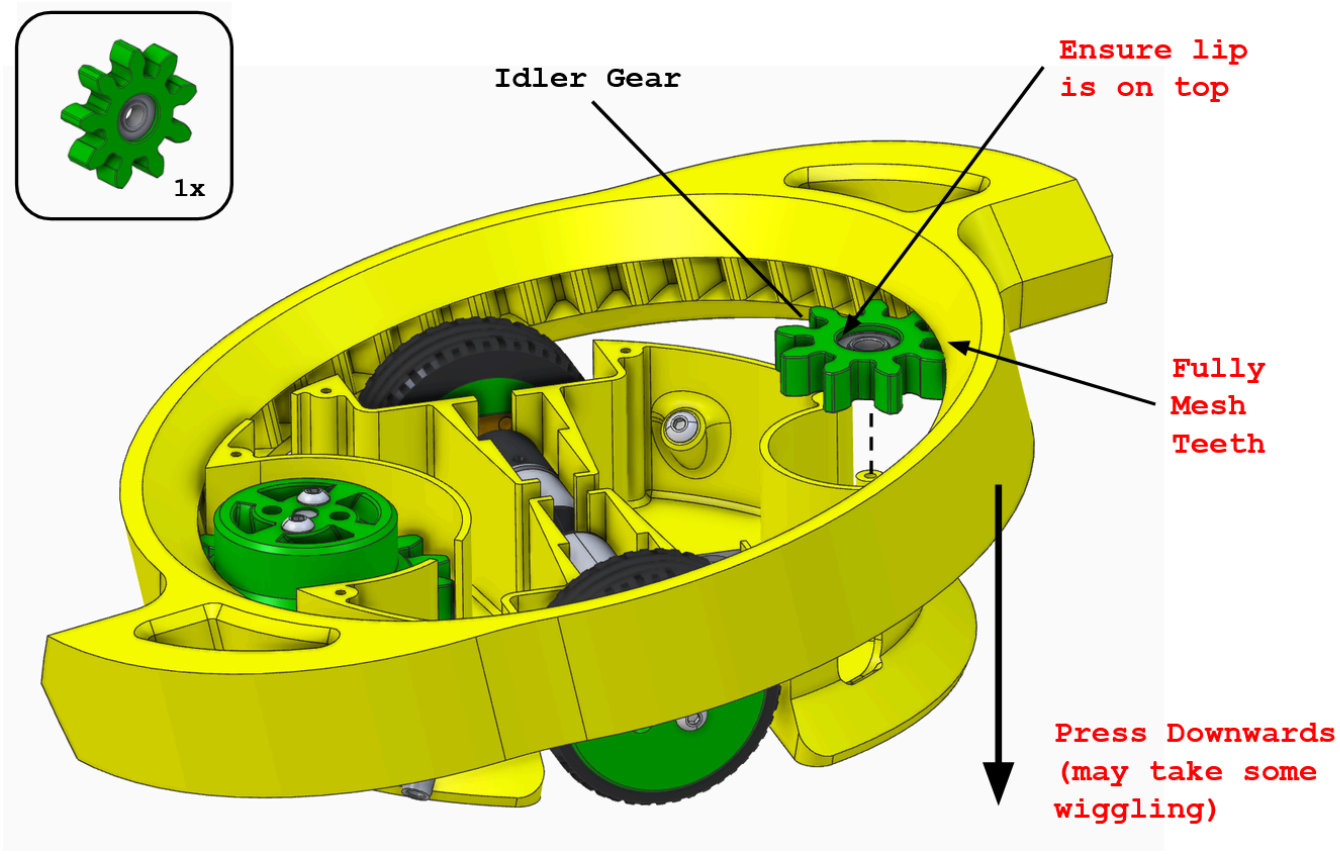


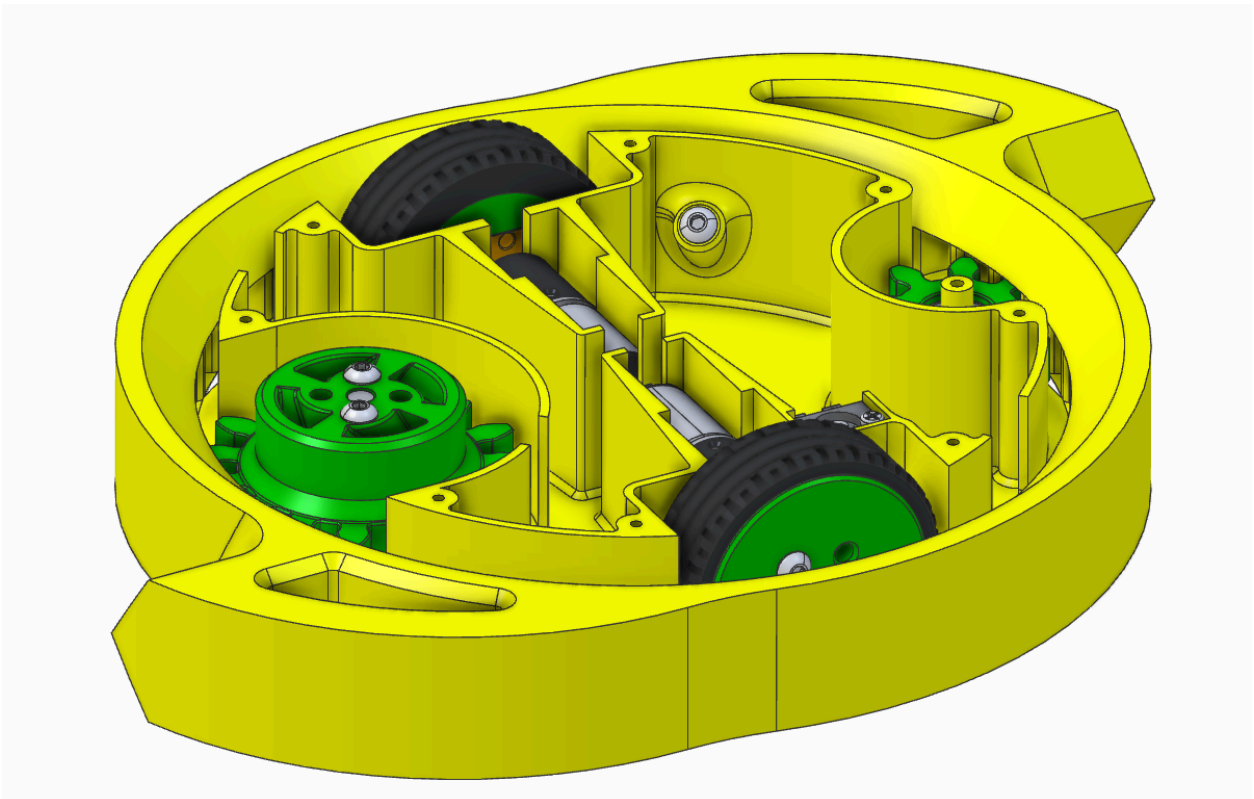
3. Mesh weapon ring teeth with drive gear and lift weapon ring rear upwards.





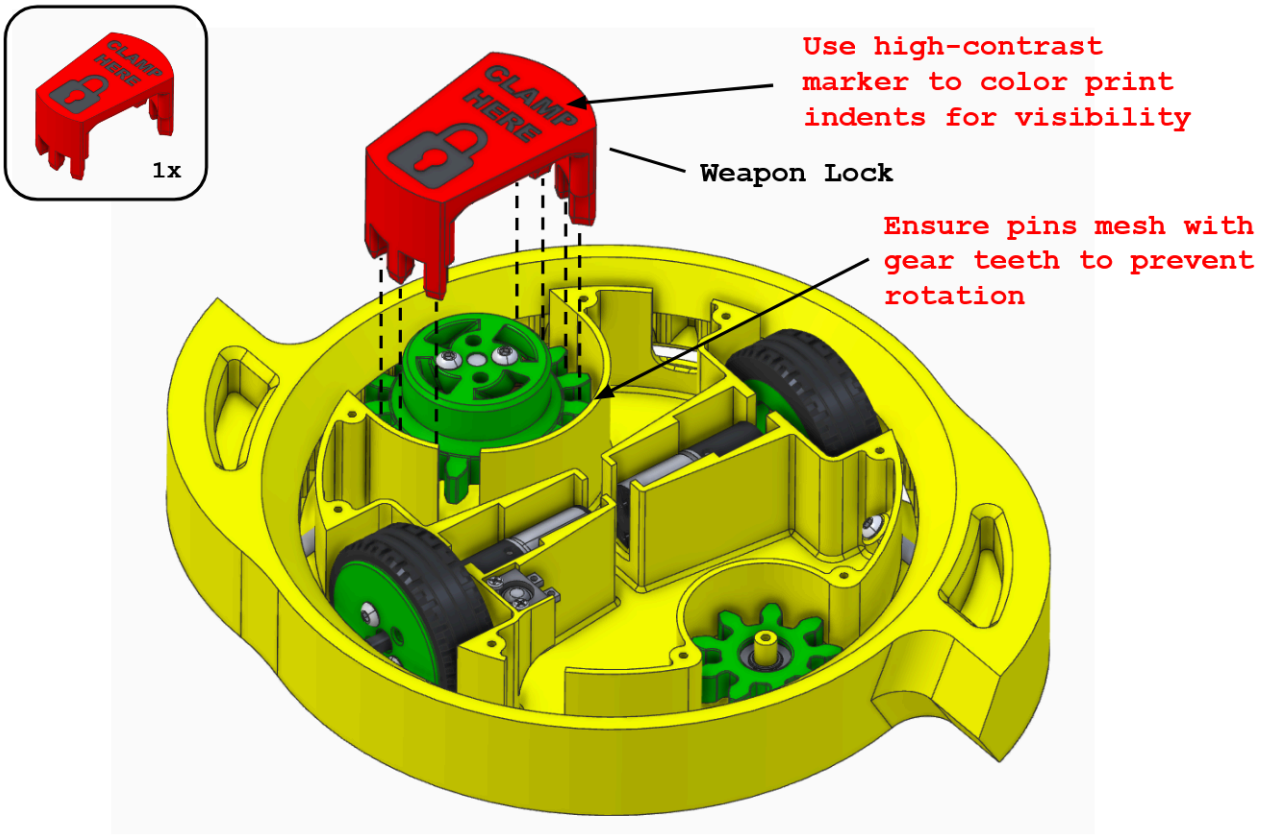
4. Mesh idler gear in rear, align with idler axle post, then press down.

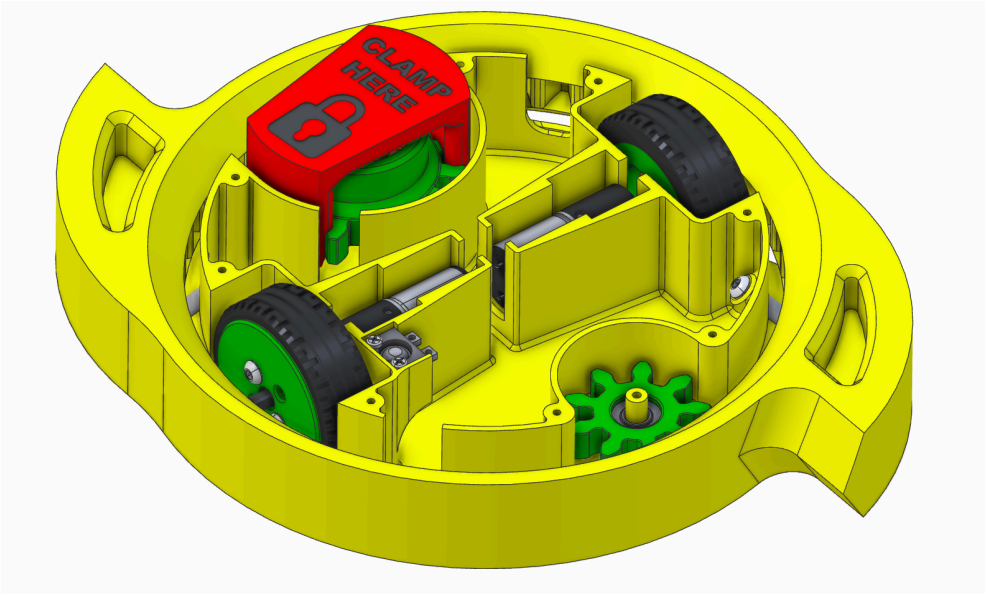




5. Install the weapon lock to prevent weapon rotation.

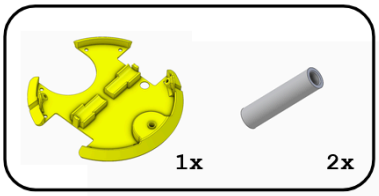
The printed weapon lock piece is NOT a sufficient weapon lock on its own, it is only designed to prevent weapon ring rotation. It should be paired with a basic clamp that can clamp the bottom of the body and the printed lock together to prevent it from coming out if the robot is turned upside down accidently.

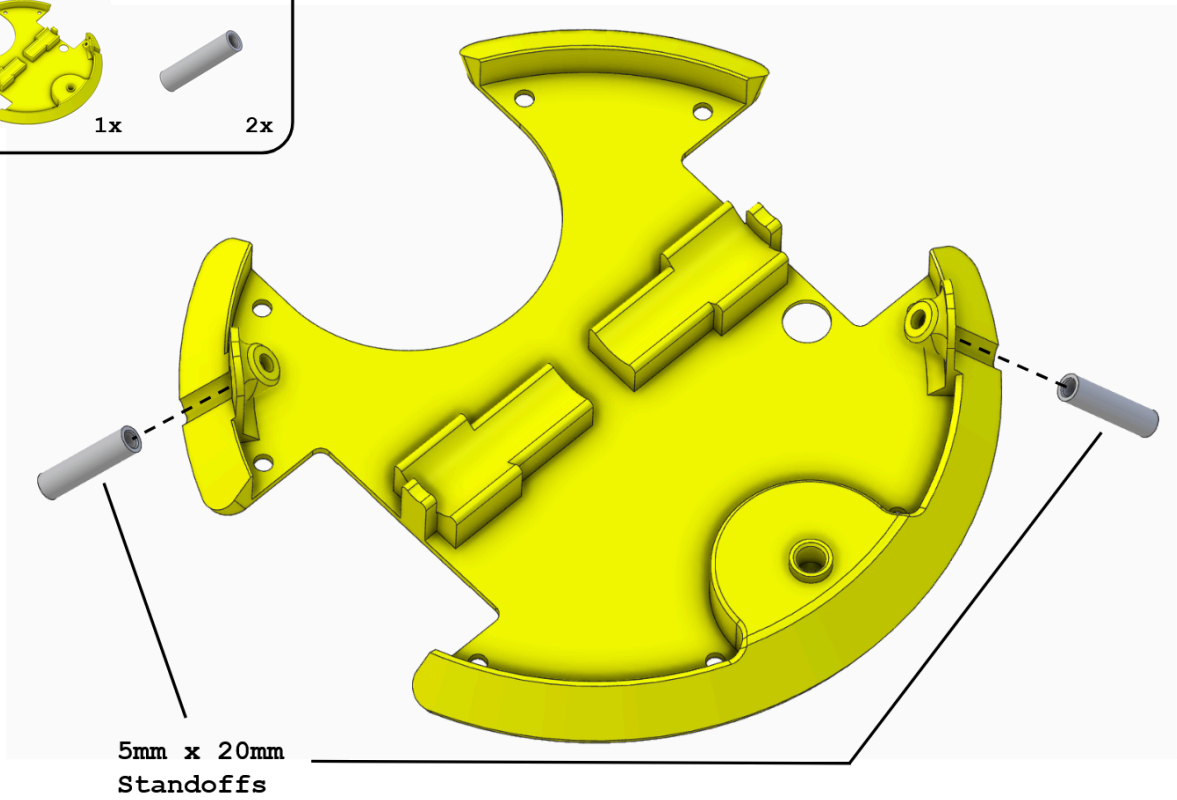


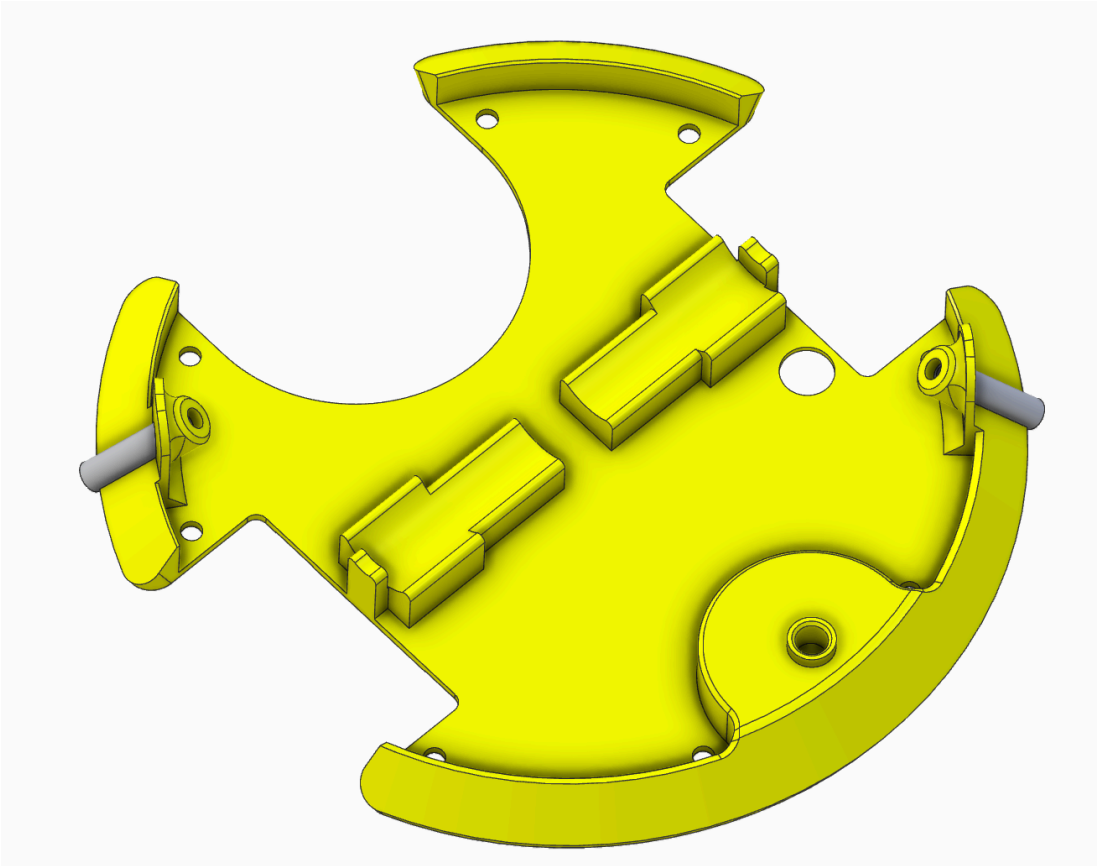


C. Upper Core Assembly

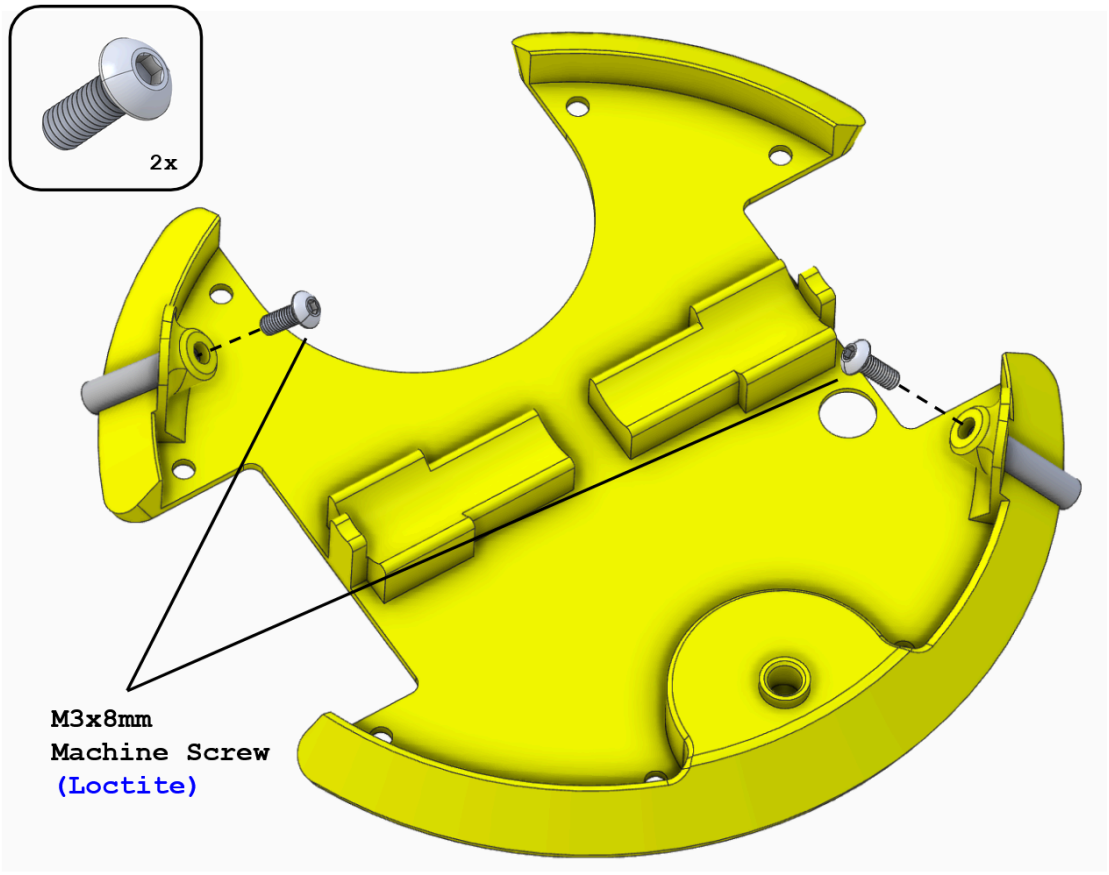
1. Insert 5mm x 20mm standoffs into locating grooves.

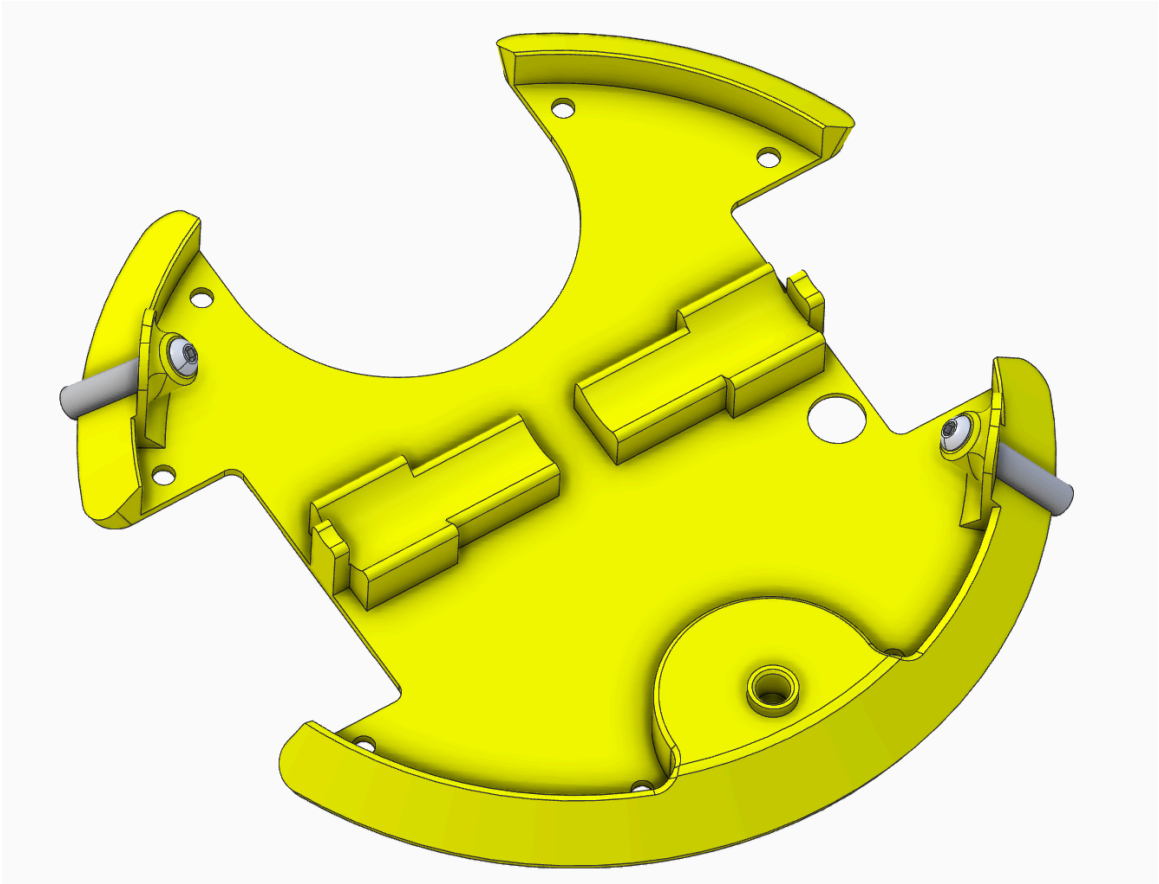






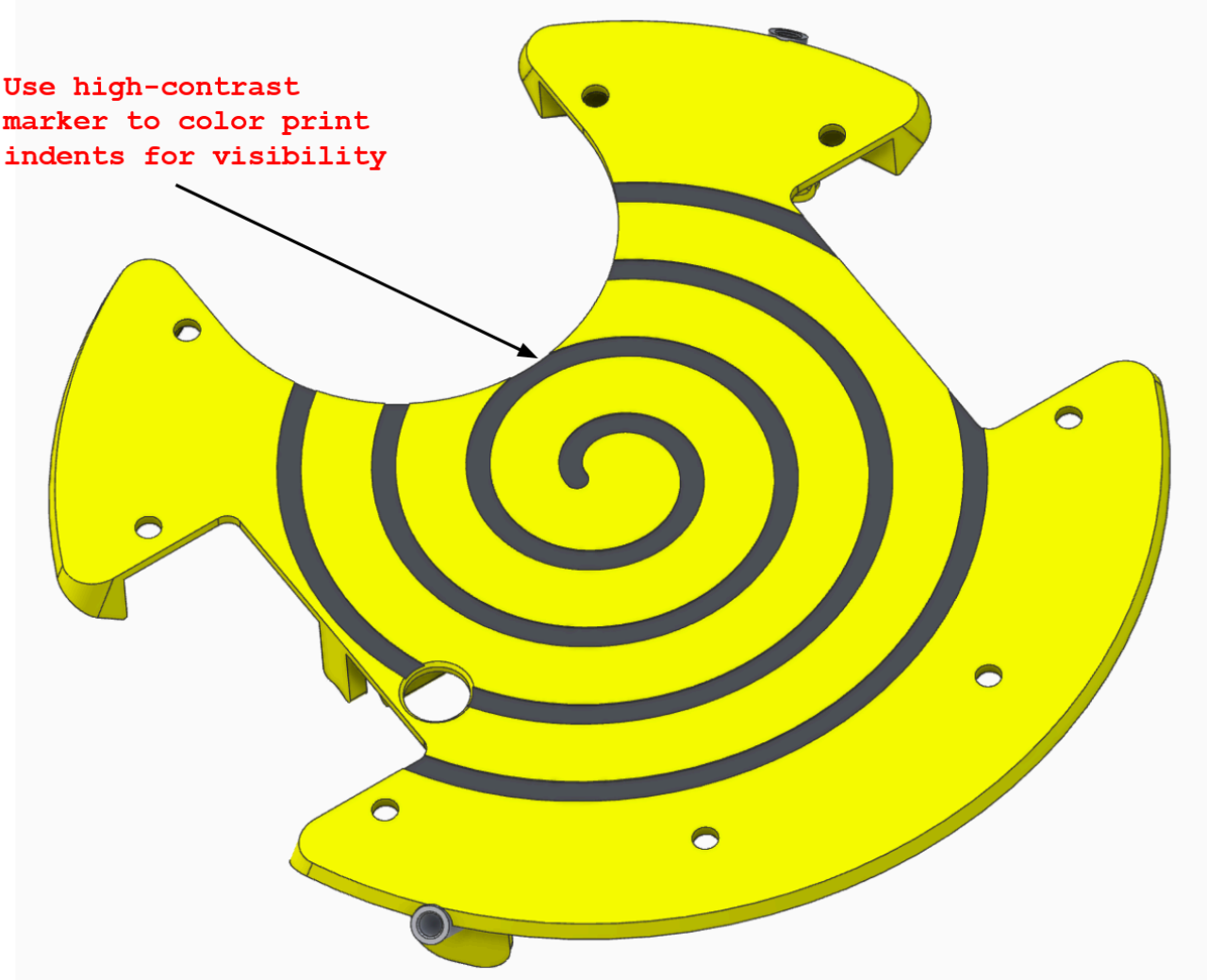
2. Attach guide standoffs with M3x8mm machine screws.





3. Flip and add accent color to top plate.

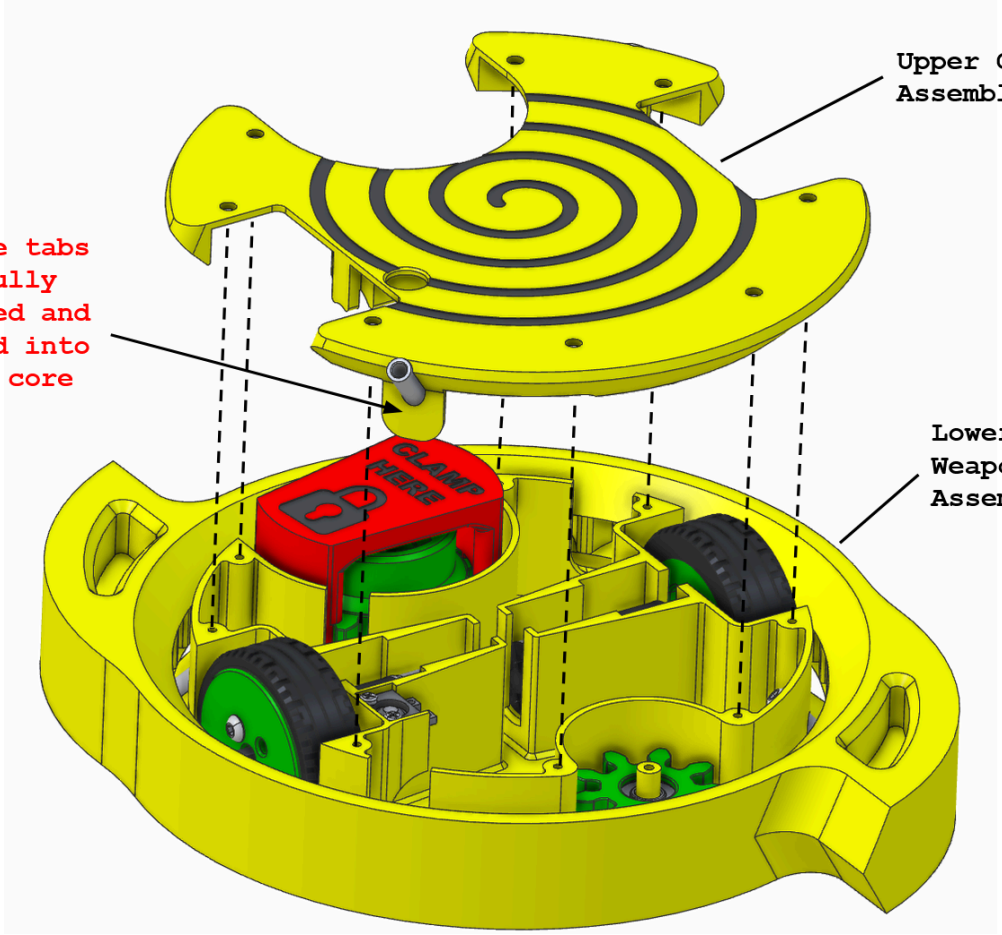
(A blank top plate file is included in the "Extras" folder to let you make your own designs!)

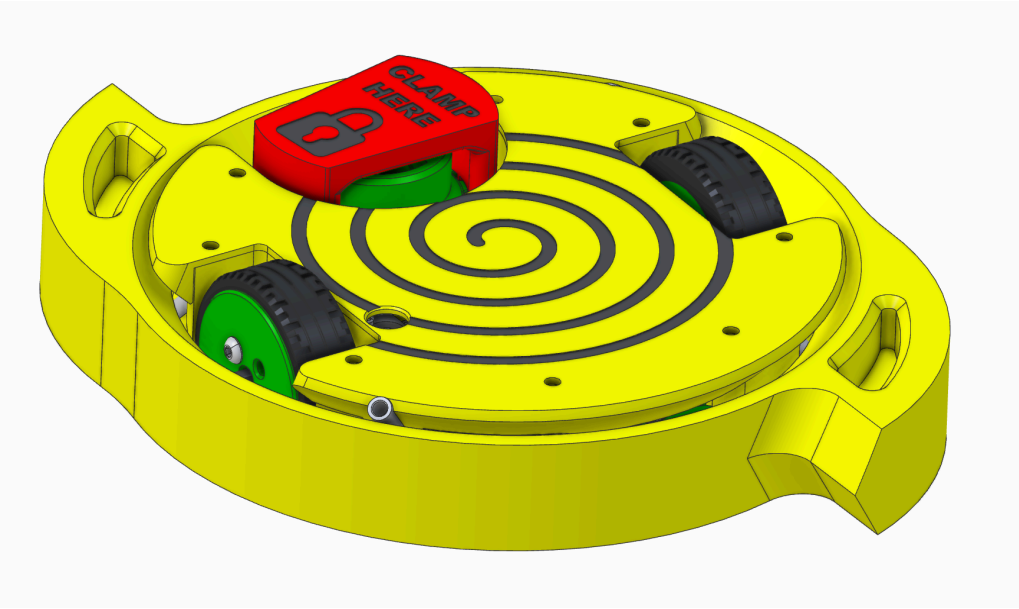


D. Final Assembly

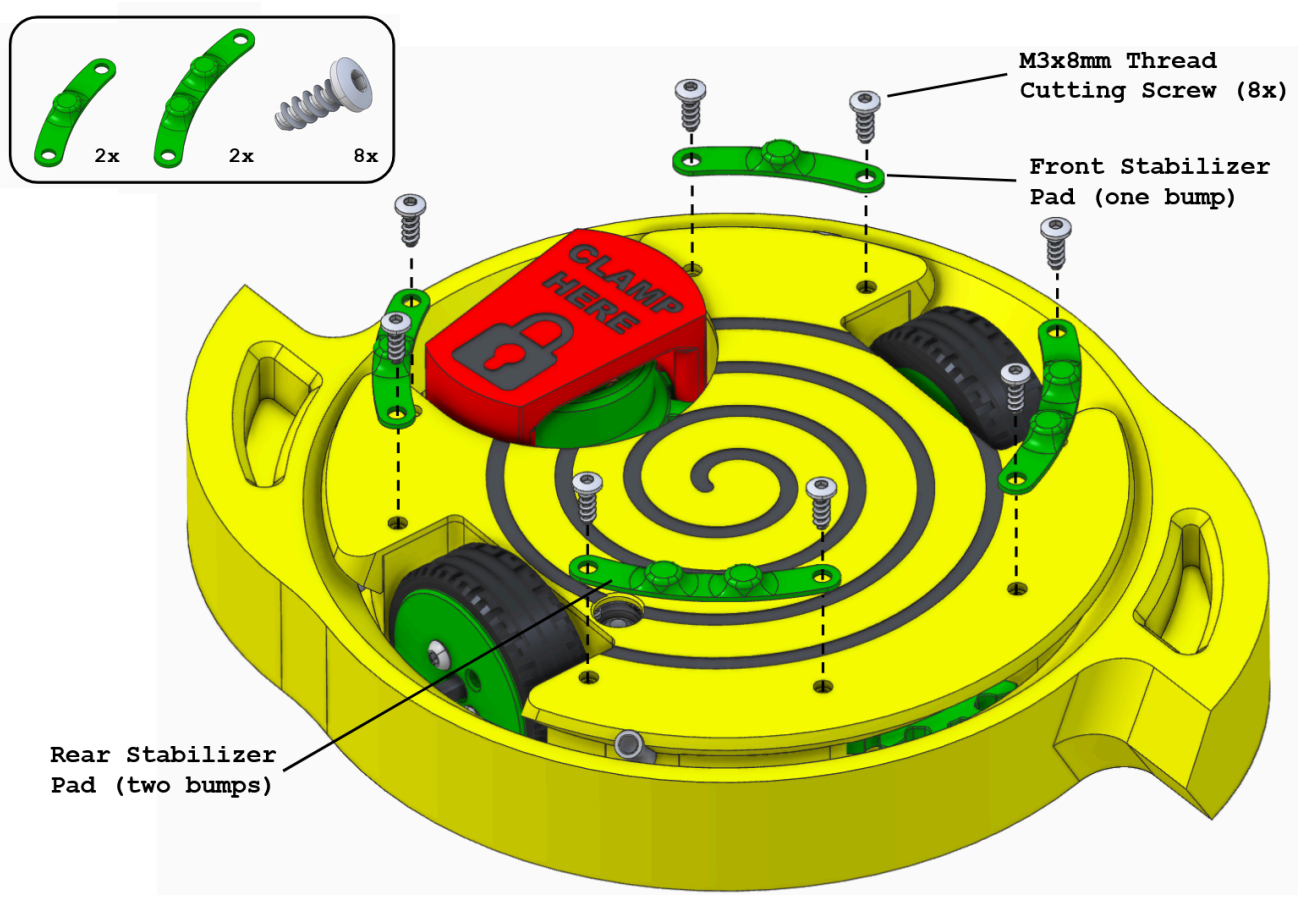
1. Align the top plate and close core.

If you're doing an assembly to run or for testing, ensure the power switch is open before installing a battery. The weapon lock and clamp should be kept installed during the final assembly for safety.





2. Attach the remaining set of stabilizer pads with M3x8mm thread cutting screws.



fin.

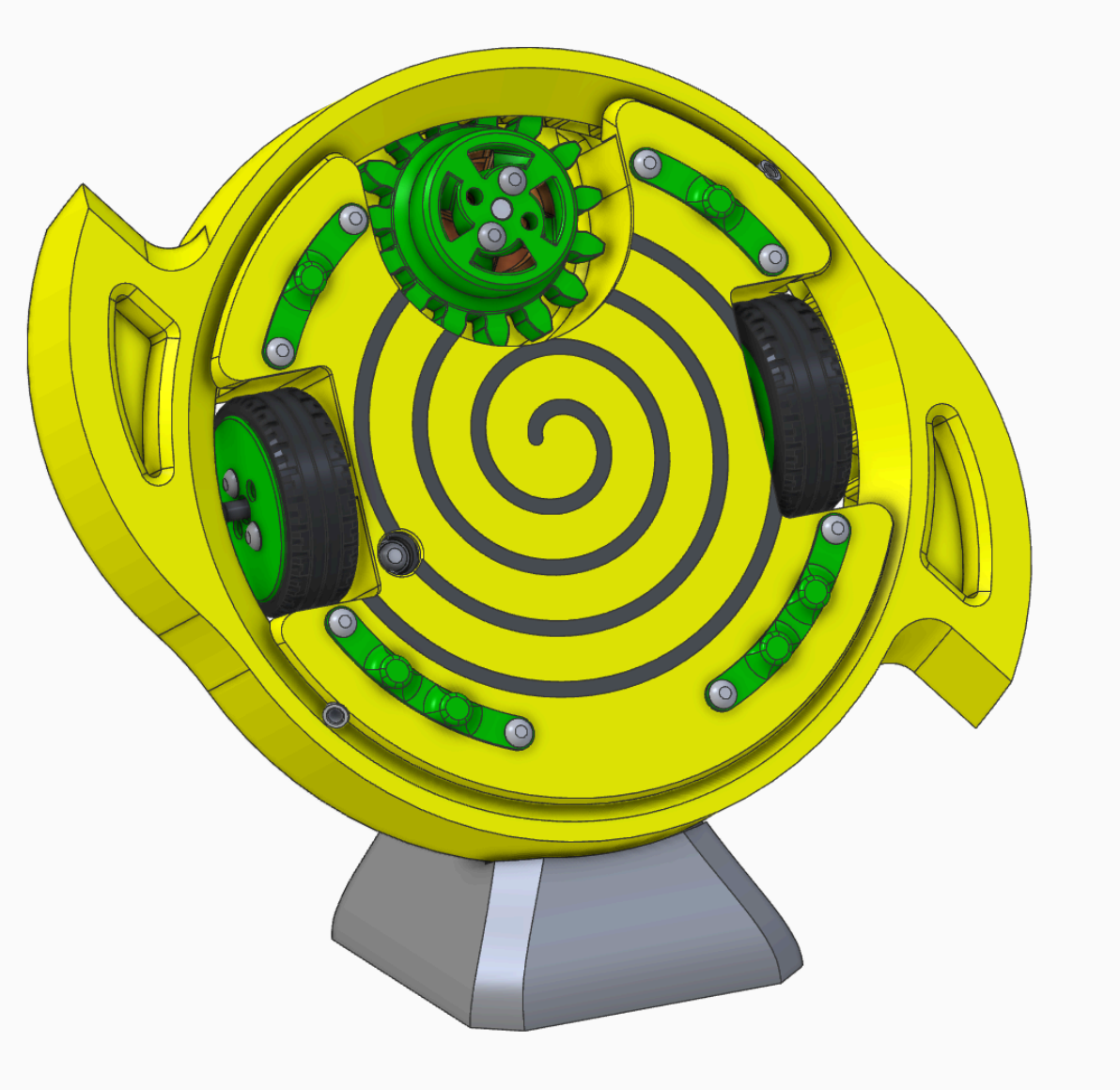
Go Fight!

E. Next Steps!

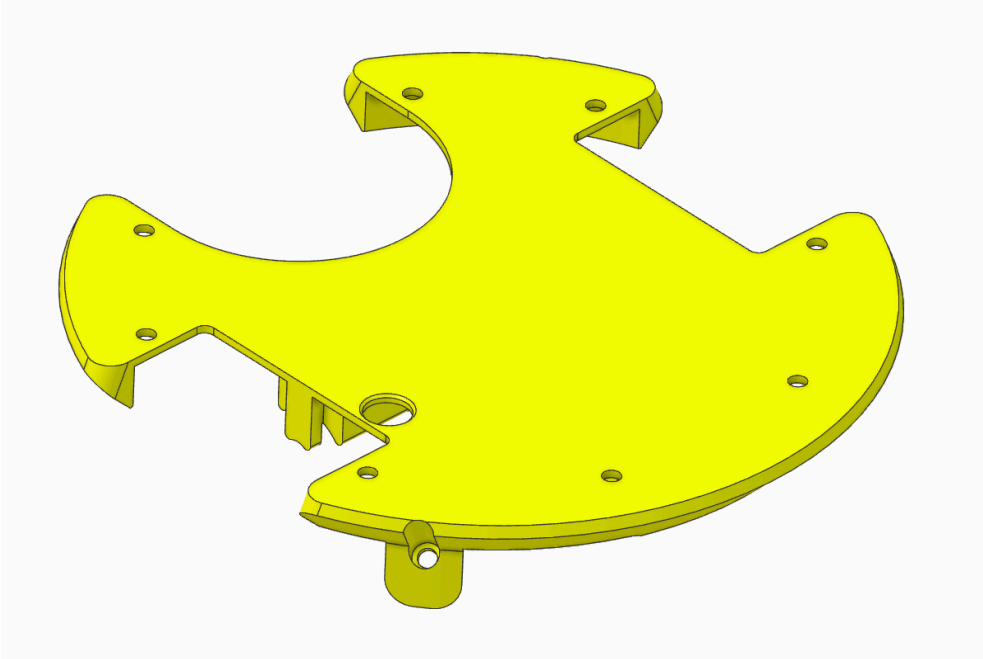
So... You built your robot, fought, and want to start exploring more. Where do you start?

Check out the "Extras" folder on the Citrus Twist GitHub page! There are three fun but easy things to do.

First, print out a display stand! This is a neat way to display your ringspinner on a shelf or desk. If you want to stop it from sliding around, add a little foam tape to the bottom of the stand.



Next, check out the upper_core_base.stp file! This is the exact same top plate, but the swirl design has been removed. Pick a CAD program of your choice, load up the part, and add a custom logo or design to make your robot truly your own. Use an extrude/cut to cut the pattern into the top plate by 0.2mm (single print layer).



Finally, open up the weapon_ring_base.stp file. This is the weapon ring, but with both of the impactor teeth removed for you to design your own! Again, load this up in a CAD program and go wild designing new teeth, impactors, and other things to spin around the outside of your robot. Just make sure that the weight is evenly distributed or things might get crazy!